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Research article

Advanced hybrid cooling approach to improve thermal management in photovoltaic thermal collectors utilizing nanofluids and nano-enhanced bio-phase change materials

Singgih Dwi Prasetyo [a,b], Zainal Arifin [c,*], Yuki Trisnoaji [b], Mohd Afzanizam Mohd Rosli [d], Ahmad Fudholi [e]

[a] Post-doctoral Researcher, Department of Mechanical Engineering, Universitas Sebelas Maret, Surakarta, 57126, Indonesia
[b] Power Plant Engineering Technology, State University of Malang, Malang, 65145, Indonesia
[c] Department of Mechanical Engineering, Universitas Sebelas Maret, Surakarta, 57126, Indonesia
[d] Department of Engineering, Faculty of Mechanical Technology and Engineering, Universiti Teknikal Malaysia Melaka, Melaka, 76110, Malaysia
[e] Research Center for Energy Conversion and Conservation, National Research and Innovation Agency, BRIN, South Tangerang, Indonesia

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ABSTRACT

Increasing thermal accumulation under high solar irradiation can degrade PV electrical performance and operational stability, highlighting the need for integrated cooling and predictive thermal management. This study aims to develop and evaluate a hybrid PV cooling framework that combines nano-enhanced bio-PCM, TiO_2/water nanofluid, and a hybrid RNN–LSTM model for thermal regulation and intelligent forecasting. A three-dimensional transient CFD model was implemented in ANSYS Fluent at solar irradiation levels of 600, 800, and 1000 W/m^2, using Palmitic Acid-TiO_2, Lauric Acid-TiO_2, Coconut Oil-TiO_2, and Beeswax-TiO_2 with 5% TiO_2/water nanofluid. The novel contribution is the integration of a sustainable, nano-enhanced bio-PCM, active nanofluid cooling, and hybrid DL prediction into a unified PV thermal management framework. The results showed that Coconut Oil-TiO_2 provided the strongest thermal regulation, achieving a minimum PV temperature of approximately 313.07 K at 600 W/m^2 and the highest thermal efficiency of 52.50% at 1000 W/m^2, while electrical efficiency remained within 17.26–18.81%. The hybrid RNN–LSTM model demonstrated reliable thermal forecasting with low residual and RMSE values. Overall, the proposed framework enhances thermal regulation, heat absorption, and predictive capability, supporting the development of smart and sustainable PV cooling systems.

1. Introduction

1.1. Background

The growing global demand for clean and sustainable energy has accelerated the deployment of PV technology across residential, commercial, and utility-scale applications. Despite its considerable potential for renewable electricity generation, PV systems remain constrained by thermal accumulation, which can adversely affect energy conversion efficiency, operational stability, and long-term reliability [1–3]. Under high solar irradiance, a substantial portion of incident solar energy is converted into heat rather than electricity, causing the PV operating temperature to increase progressively. Elevated PV temperature can reduce electrical efficiency, accelerate material degradation, increase thermal stress, and ultimately shorten the service life of PV modules. These challenges are particularly important in tropical and subtropical environments, where high ambient temperatures and intense solar exposure can intensify thermal accumulation during peak operating periods [4,5]. Consequently, effective thermal regulation has become an important research direction for maintaining favorable PV operating temperatures, improving energy conversion stability, and supporting the long-term deployment of sustainable PV systems [6,7].

Among the available thermal management approaches, PCM-based cooling has attracted considerable research interest because of its passive operation and latent heat storage capability [8,9]. During the melting process, the PCM absorbs a substantial amount of thermal energy while maintaining a relatively stable temperature, thereby

* Corresponding author.
E-mail address: zainal_arifin@staff.uns.ac.id (Z. Arifin).

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Nomenclature and abbreviations

PV	Photovoltaic
PVT	Photovoltaic Thermal
PCM	Phase Change Material
Bio-PCM	Bio-based Phase Change Material
NePCM	Nano-enhanced Phase Change Material
CFD	Computational Fluid Dynamics
DL	Deep Learning
AI	Artificial Intelligence
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
ANOVA	Analysis of Variance
FEM	Finite Element Method
FEA	Finite Element Analysis
CNT	Carbon Nanotube
SWCNT	Single-Walled Carbon Nanotube
GO	Graphene Oxide
TEG	Thermoelectric Generator
EVA	Ethylene Vinyl Acetate
PVF	Polyvinyl Fluoride
TiO_2	Titanium Dioxide Nanoparticle
Al_2O_3	Aluminum Oxide Nanoparticle
CuO	Copper Oxide Nanoparticle
MgO	Magnesium Oxide Nanoparticle
RT-42	Commercial PCM Material
MAPE	Mean Absolute Percentage Error, %
MSE	Mean Squared Error
RMSE	Root Mean Square Error, K
MAE	Mean Absolute Error, K
SIMPLE	Semi-Implicit Method for Pressure-Linked Equations
PRESTO	Pressure Staggering Option
Q	Total absorbed thermal energy, J
m	Mass, kg
C_p	Specific heat capacity, J/kg·K
ΔT	Temperature difference, K
L	Latent heat, J/kg
k_{eff}	Effective thermal conductivity, W/m·K
k_f	Thermal conductivity of base fluid, W/m·K
k_p	Thermal conductivity of a nanoparticle, W/m·K
ϕ	Nanoparticle volume fraction, %
q	Convective heat transfer rate, W
h	Convection heat transfer coefficient, W/m²·K
A	Heat transfer surface area, m²
T_s	Surface temperature, K
T_∞	Ambient temperature, K
q_s	Solar heat flux, W/m²
Q_{abs}	Absorbed thermal energy, W
η_e	Electrical efficiency, %
η_{ref}	Reference electrical efficiency, %
β	Temperature coefficient, K^{-1}
T_{pv}	PV operating temperature, K
T_{ref}	Reference temperature, K
η_{th}	Thermal efficiency, %
G	Solar irradiation intensity, W/m²
t	Simulation time, s
X_{norm}	Normalized data
X	Original data value
$X_{\min}$	Minimum dataset value
$X_{\max}$	Maximum dataset value
h_t	Hidden state at time step (t)
x_t	Input data at time step (t)
W	Weight matrix
b	Bias parameter
tanh	Hyperbolic tangent activation function
f_t	Forget gate
C_t	Cell state memory
i_t	Input gate
$\widetilde{C}_t$	Candidate memory state
y_i	Actual value
$\hat{y}_i$	Predicted value
SS	Sum of Squares
df	Degrees of Freedom
MS	Mean Square
F	F-statistic
$P-value$	Probability value
(F_{crit})	Critical F-value
K	Kelvin, K
W/m²	Watt per square meter
J/kg	Joule per kilogram
J/kg·K	Joule per kilogram Kelvin
W/m·K	Watt per meter Kelvin
kg/m³	Kilogram per cubic meter
kg/m·s	Kilogram per meter second
kWh/m²·year	Kilowatt-hour per square meter per year
m³ /year	Cubic meter per year

reducing excessive temperature rise within the PV module. The stored thermal energy can subsequently be released during solidification, allowing the PCM to act as a buffer against transient thermal fluctuations [10,11]. However, conventional PCM materials often exhibit relatively low thermal conductivity, which can limit heat penetration into the storage medium and reduce the rate of heat absorption. This limitation becomes more pronounced under prolonged solar irradiation, when continuous heat generation may exceed the PCM layer's effective heat-transfer capacity. Therefore, nano-enhanced PCMs have increasingly been investigated by incorporating nanoparticles to improve thermal conductivity, thermal diffusion, melting behavior, and temperature uniformity. In parallel, bio-PCM has emerged as a sustainable alternative because of its favorable environmental characteristics, biodegradability, relatively low toxicity, and potential to reduce dependence on petroleum-derived thermal storage materials [12–14]. The combination of bio-PCM and nanoparticle enhancement consequently provides an attractive pathway for simultaneously improving PV thermal regulation and the sustainability characteristics of thermal energy storage.

In addition to PCM-based thermal storage, active cooling using nanofluids has become an important approach for enhancing PV heat dissipation [15–17]. Nanofluids contain suspended nanoparticles that can modify the thermophysical characteristics of conventional base fluids and improve thermal conductivity, thermal diffusivity, convective heat transfer, and overall heat transport capability. Previous investigations have reported the use of nanoparticles such as Al_2O_3, CuO, TiO_2, CNT, and graphene derivatives to improve heat removal and temperature uniformity in thermal energy and PV applications [18,19]. Compared with conventional water-based cooling, nanofluids can provide enhanced heat-transfer performance and faster thermal response, particularly when heat generation varies with solar irradiance. However, active nanofluid cooling alone requires continuous fluid circulation and may not fully address transient thermal storage requirements during periods of high solar loading. The integration of nanofluid cooling with nano-enhanced bio-PCM, therefore, provides a complementary thermal management mechanism in which active convective

heat dissipation can operate together with passive latent heat storage [20,21]. Such a combination can potentially improve heat removal during active cooling while simultaneously buffering excessive thermal energy within the PCM layer. Therefore, hybrid configurations combining nano-enhanced bio-PCM and nanofluid cooling represent a promising direction for developing more effective PV thermal management systems under variable solar operating conditions.

Beyond the development of thermal materials and cooling architectures, the increasing complexity of PV operating conditions has created a need for predictive and intelligent thermal management. Conventional cooling approaches generally operate under predetermined thermal configurations and therefore have limited ability to anticipate changes driven by fluctuating solar irradiance, ambient temperature, heat generation, and transient thermal loading. The integration of AI and DL into renewable energy systems has consequently received increasing attention because predictive models can identify nonlinear relationships within operational datasets and support more adaptive system management [22–24]. Among DL techniques, RNN–LSTM architectures are particularly relevant to thermal–electrical systems because sequential operating data contain temporal dependencies that conventional static models cannot always adequately represent [25,26]. The combination of RNN and LSTM structures can facilitate the representation of both short-term variations and longer-term dependencies in thermal and electrical responses. Such predictive capability can complement CFD analysis by transforming simulation-generated thermal–electrical data into a forecasting framework for dynamic operating conditions. Accordingly, integrating intelligent prediction with advanced cooling technologies offers an opportunity to move PV thermal management beyond conventional passive or active cooling toward adaptive, data-driven operation. This research direction is particularly relevant for next-generation PV systems that require not only improved thermal performance but also reliable prediction of operating behavior under changing environmental conditions.

1.2. Research gap, novelty, aim, and objectives

Recent studies have demonstrated substantial progress in PV thermal management through PCM, nanofluids, fins, heatsinks, reflectors, and CFD-based optimization, as summarized in Table 1. These studies have confirmed that passive and active cooling strategies can improve PV thermal behavior and, consequently, influence electrical and thermal performance. Nevertheless, the existing literature remains fragmented because many investigations have focused on either PCM enhancement or nanofluid cooling, rather than on integrating the two within a single thermal management architecture. Furthermore, the application of nano-enhanced bio-PCMs in combination with nanoparticle-based cooling fluids remains relatively limited, particularly when the objective is to simultaneously exploit latent heat storage and active convective heat dissipation. Another important limitation is that most CFD-based studies primarily evaluate temperature distribution, melting behavior, and efficiency enhancement without integrating predictive models capable of learning the temporal characteristics of PV thermal–electrical behavior. Consequently, conventional cooling configurations remain predominantly reactive rather than predictive, limiting their adaptability under dynamically changing solar irradiation and thermal loading conditions. The integration of advanced sustainable thermal storage, active nanofluid cooling, and DL-based forecasting, therefore, represents an important unresolved research direction in PV thermal management. Addressing this gap requires a unified framework capable of evaluating thermal regulation, energy conversion, and predictive performance within the same system.

The primary novelty of this study is the development of an integrated hybrid PV cooling framework that combines nano-enhanced bio-PCM, TiO_2/water nanofluid cooling, and a hybrid RNN–LSTM prediction model within a unified thermal management architecture. The study aims to investigate how the combined cooling configuration influences PV thermal regulation, melting behavior, thermal efficiency, electrical efficiency, and predictive performance under different solar irradiation conditions. Specifically, the objectives of this study are as follows: **(1)** to evaluate the transient thermal response of Palmitic Acid-TiO_2, Lauric Acid-TiO_2, Coconut Oil-TiO_2, and Beeswax-TiO_2 under irradiation intensities of 600, 800, and 1000 W/m^2 using three-dimensional transient CFD simulations; **(2)** to determine the influence of irradiation intensity and material configuration on PV temperature, PCM melting behavior, thermal efficiency, and electrical efficiency, supported by statistical evaluation using ANOVA; **(3)** to develop and evaluate a hybrid RNN–LSTM model using CFD-generated thermal–electrical datasets to predict the transient thermal behavior of the proposed system; and **(4)** to assess the broader sustainability implications of the predicted cooling performance in terms of energy and environmental benefits. Through these objectives, the study contributes an integrated numerical and intelligent framework that connects sustainable thermal storage, active heat dissipation, and predictive DL. This framework provides a comprehensive basis for advancing adaptive, reliable, and sustainable PV cooling systems under variable solar operating conditions.

2. Methodology

2.1. Research framework and numerical simulation procedure

This study employed a systematic numerical methodology to investigate the thermal–electrical performance of a hybrid PV cooling system integrating nano-enhanced bio-PCM, nanofluid cooling, and a hybrid RNN–LSTM predictive model. The overall research workflow consisted of several sequential stages: literature review, data collection, thermal system design, CFD simulation, validation, and intelligent performance analysis. Initially, a comprehensive literature review was conducted to identify the most relevant thermal management approaches, nanoparticle materials, and intelligent prediction techniques for advanced PV cooling applications. The collected data included the thermophysical properties of the nano-enhanced bio-PCM, the TiO_2/water nanofluid characteristics, the PV operating parameters, and the environmental boundary conditions under varying solar irradiation levels. The designed cooling configuration incorporated a nanoparticle-enhanced bio-PCM combined with a TiO_2/water nanofluid at 5% concentration to simultaneously improve latent heat storage capability and convective heat transfer performance. Furthermore, several operating conditions, including solar irradiation intensity, ambient temperature (300 K), convection coefficient, and mass flow rate, were used to evaluate the thermal response of the proposed system under different environmental conditions.

The overall numerical workflow and simulation procedure adopted in this study are illustrated in Fig. 1. Before conducting the main CFD simulation, validation and mesh independence tests were performed to ensure numerical stability, computational accuracy, and the reliability of the results. The validation criterion used MAPE below 10% as the primary acceptance threshold for simulation feasibility and consistency. If the numerical results did not meet the predefined validation criterion, the simulation configuration and mesh structure were iteratively refined until acceptable agreement was achieved. After successful validation, comprehensive CFD simulations were conducted using ANSYS to analyze temperature distribution, heat dissipation characteristics, thermal storage behavior, and PV thermal–electrical performance. Subsequently, ANOVA and hybrid RNN–LSTM algorithms were integrated to evaluate the influence of thermal parameters and to develop intelligent predictive capability for dynamic PV operating conditions. The incorporation of hybrid DL modeling enabled adaptive forecasting of thermal–electrical behavior, thereby improving prediction accuracy, enabling intelligent thermal management, and enhancing the operational reliability of the proposed next-generation PV cooling framework.

Table 1
Previous studies on PV thermal management using PCM nanofluid and hybrid cooling technologies.

Reference	Focus Study	Method	Electrical Efficiency	Thermal Efficiency	Key Result
A. S. Abdelrazik et al. (2023) [27]	Comprehensive review of ANSYS-Fluent CFD applications in solar thermal, PV/T, and PV/PCM systems	Review study using CFD-based simulation analysis and validation of solar systems	PV efficiency enhanced by 33–46% with PCM integration	Effective cooling reduced PV temperature by 15–20%	The review highlighted that CFD simulation is an effective optimization tool for solar energy systems. Nanofluids significantly improved thermal performance, while PCM integration enhanced PV efficiency and extended thermal management duration.
Hiwa Abdlla Maarof et al. (2025) [28]	Optimization of double-pass PVT system integrated with PCM and wavy fins (PVT-PCMWF)	3D CFD simulation using COMSOL Multiphysics	12.09%	18%	Integration of wavy fins increased turbulent airflow and heat transfer, improving thermal efficiency by 61.35% and daily heat gain by 72% compared to conventional PVT-PCM systems.
T. Usha Rani et al. (2025) [29]	Investigation of GO/water nanofluid effects on PVT performance	CFD simulation coupled with ANOVA statistical optimization	Increased from 9.2% to 11.8%	Increased from 66% to 80%	Incorporation of 0.5 wt% GO nanofluid substantially enhanced thermal and electrical efficiencies, achieving an overall energy efficiency of 83.5% and exergy efficiency of 18.3%.
Raman Kumar et al. (2025) [30]	Adaptive reflector-assisted PVT system integrated with PCM thermal buffering	Transient CFD–FEA multiphysics simulation	Combined energy conversion improved significantly	Combined energy efficiency reached 72.5%	The integration of the adaptive reflector and PCM stabilized PV temperature near the PCM melting point, increased exergy efficiency by 14.7%, and substantially reduced entropy generation.
Ahmad Alqatamin & Jinzhan Su (2025) [31]	Enhancement of PVT performance using perforated V-shaped heatsink and nanofluids	3D CFD simulation in ANSYS Fluent validated experimentally	Up to 17%	Up to 83.5%	A CNT/water nanofluid with an optimized heatsink geometry reduced PV temperature by 7.9 K and improved thermal efficiency by approximately 19% at higher nanoparticle concentrations.
Hui Chen et al. (2024) [32]	Investigation of intermittent nanofluid cooling in PVT/NPCM systems	2D transient numerical heat transfer model using FEM	Average of 19.7%	Maintained comparable thermal performance	Intermittent cooling significantly reduced pumping energy consumption while maintaining electrical efficiency nearly identical to that of continuous cooling systems.
Hiwa Abdlla Maarof et al. (2025) [33]	Bi-fluid PVT system integrated with perforated fins and PCM (Bi-PVT-PCM-PF)	3D CFD multiphysics simulation in COMSOL validated experimentally	Increased from 11.98% to 12.97%	Heat dissipation enhanced by 28.3%	A combination of bi-fluid cooling, PCM, and perforated fins reduced PV temperature by 16.1 °C and improved daily energy output by 9.2%.
Ali Gholami et al. (2025) [34]	Hybrid PCM enhanced with Al2O3 and MgO nanoparticles for PV cooling	Numerical analysis of PV/HPCM system with fins and heatsinks	Increased from 21.15% to 24%	Improved due to enhanced heat transfer	Hybrid nanoparticles accelerated PCM melting and significantly improved PV efficiency, especially at higher nanoparticle concentrations.
Congchao Zeng et al. (2025) [35]	PV/T-PCM system integrated with bionic fractal fins	3D transient CFD simulation	Improved by approximately 5%	Increased from 54.96% to 80.36%	Bionic fractal fins increased the heat-transfer area and thermal uniformity, significantly reducing PV temperature and improving both thermal and electrical performance.
Mohammed El Goual et al. (2026) [36]	Passive finned PV-PCM system with multi-layer PCM configuration	Numerical optimization of fin geometry and PCM arrangement	Increased by 8.21%	Enhanced thermal regulation performance	Optimized fin thickness and dual-PCM arrangement reduced PV temperature by 16.29 °C and significantly improved passive cooling efficiency.
A. M. Alinia & M. Sheikholeslami (2025) [37]	PV system integrated with a parabolic reflector, PCM cooling, SWCNT nanoparticles, and TEG	3D transient CFD simulation using ANSYS Fluent and SolTrace	PV electrical efficiency improved by 16.46%	Thermal efficiency increased by 8.47%	The combined integration of reflectors, nanocomposite PCM cooling, fins, and TEG increased electrical output power by up to 2.94 times compared to conventional PV systems.
This Study	Numerical CFD simulation of PV cooling using nanoparticle-enhanced bio-PCM supported by nanoparticle-based cooling fluid integrated with deep learning prediction	3D CFD simulation in ANSYS Fluent coupled with a hybrid RNN–LSTM deep learning model	Expected to improve electrical efficiency through enhanced thermal regulation	Expected to enhance thermal energy storage and heat dissipation performance	This study proposes an advanced hybrid framework that combines a nano-enhanced bio-PCM with nanoparticle-based fluid cooling to reduce PV operating temperature and improve energy conversion performance. In addition, a hybrid RNN–LSTM deep learning model is incorporated to predict and optimize thermal–electrical behavior, enabling more accurate performance forecasting, intelligent thermal management, and enhanced reliability of next-generation PV cooling systems.

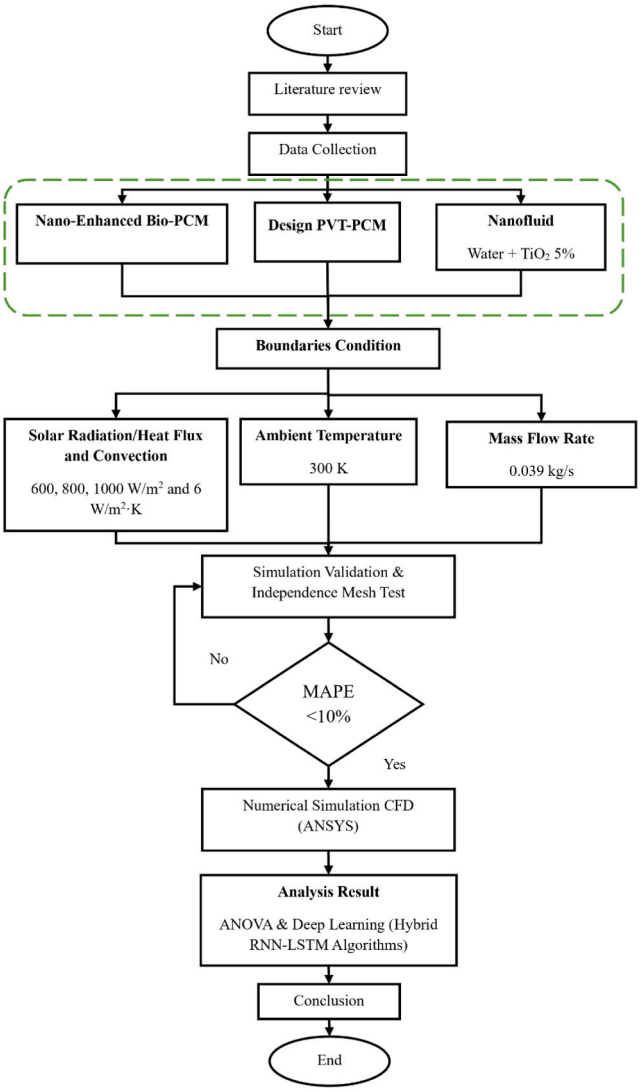


Fig. 1. Research framework and numerical simulation procedure of a hybrid PV cooling system integrated with nano-enhanced bio-PCM nanofluid and a hybrid RNN–LSTM Prediction model.

2.2. Geometrical design and material configuration of the hybrid PV cooling system

The proposed hybrid PV cooling system was designed using a multilayer thermal configuration integrating conventional PV structural components, nanofluid cooling channels, and nano-enhanced bio-PCM thermal storage media. The detailed geometrical arrangement of the integrated cooling structure is illustrated in Fig. 2**(a), which shows** the layered architecture of the PV module combined with the thermal management system. The multilayer configuration consisted of several main sections, including the transparent protective layer, PV conversion layer, nanofluid cooling region, nano-enhanced bio-PCM thermal storage zone, and aluminum-based collector structure. The upper protective layer maximized the transmission of solar irradiance to the PV surface while simultaneously protecting the module from environmental exposure. In addition, the integrated thermal storage region beneath the PV structure temporarily absorbed excess thermal energy during high-temperature operation. The incorporation of nanofluid-circulation channels further improved heat-dissipation capability by accelerating convective heat transfer and enhancing temperature uniformity across the PV surface.

The detailed thermophysical properties and dimensional

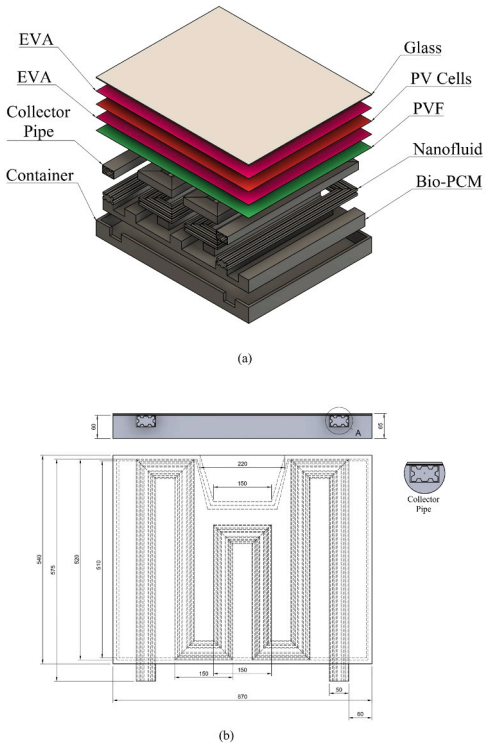


Fig. 2. Geometrical configuration of the hybrid PV cooling system: (a) Layered structure of the integrated PV module and (b) Dimensional design and flow channel geometry.

specifications of all system components are summarized in Table 2. The PV cooling architecture used aluminum-based collector pipes and PCM containers, which benefit from aluminum's high thermal conductivity, enabling rapid thermal transport and efficient heat removal from the PV module. Furthermore, the nano-enhanced bio-PCM layer provided latent heat storage capability to stabilize PV operating temperature and reduce thermal stress accumulation during continuous solar exposure. The TiO_2/water nanofluid cooling channel was designed to improve convective heat transfer performance and maintain stable thermal regulation under varying environmental conditions. As shown in Fig. 2 (b), the dimensional configuration and flow channel geometry of the proposed cooling system were carefully optimized to ensure effective interaction between the nanofluid circulation and the PCM thermal storage regions. Therefore, the integrated multilayer configuration enabled simultaneous passive and active cooling, improving thermal regulation, increasing electrical efficiency, and enhancing operational reliability for advanced PV thermal management applications.

2.3. Thermophysical properties of nano-enhanced Bio-PCM and nanofluid

The thermophysical properties of the nano-enhanced bio-PCM and nanofluid utilized in this study are summarized in Table 3. Several bio-based PCM materials, including palmitic acid, lauric acid, coconut oil, and beeswax, were enhanced with 5% TiO_2 nanoparticles to improve thermal conductivity and heat transfer. The incorporation of TiO_2 nanoparticles significantly enhanced the PCM's thermal response by accelerating heat diffusion and improving thermal uniformity during melting and solidification. Among the investigated materials, beeswax-TiO_2 exhibited the highest latent heat storage capacity of 180000 J/kg, indicating a strong ability to absorb thermal energy under high-temperature operating conditions. Meanwhile, palmitic acid-TiO_2 exhibited a relatively higher thermal conductivity in the liquid phase, which contributed to faster thermal dissipation and more effective heat

Table 2
Thermophysical properties and geometrical dimensions of the hybrid PV cooling system components [38,39].

Layer / Component	Material	Density (kg/m³)	Specific Heat (J/kg·K)	Thermal Conductivity (W/m·K)	Thickness / Dimension
Top Cover	Glass	2450	790	0.70	3.2 mm
Encapsulant Layer 1	EVA (1)	960	2090	0.311	0.5 mm
Photovoltaic Layer	PV Cells	2330	677	130	0.21 mm
Encapsulant Layer 2	EVA (2)	960	2090	0.311	0.5 mm
Back Sheet	PVF	1200	1250	0.15	0.3 mm
Nanofluid Collector Pipe	Aluminum	2719	871	202.4	50 mm × 30 mm
Nano-enhanced bio-PCM Container	Aluminum Container	2719	871	202.4	670 mm × 540 mm × 50 mm

Table 3
Thermophysical properties of nano-enhanced bio-PCM and TiO_2/water nanofluid used in the hybrid PV cooling system [40–44].

Material	φ (%)	Phase	Density (kg/m³)	Cp (J/kg·K)	Thermal Conductivity (W/m·K)	Viscosity (kg/m·s)	Latent Heat (J/kg)	Temperature (K)
Palmitic Acid-TiO_2	5	Solidus	885	2150	0.36	-	-	303
		Liquidus	900	2400	0.52	0.0065	170000	312
Lauric Acid- TiO_2	5	Solidus	905	2050	0.32	-	-	309
		Liquidus	920	2300	0.52	0.006	160000	319
Coconut Oil- TiO_2	5	Solidus	890	2100	0.33	-	-	300
		Liquidus	910	2350	0.5	0.0065	150000	310
Beeswax-TiO_2	5	Solidus	860	1950	0.34	-	-	306
		Liquidus	880	2200	0.55	0.007	180000	316
Water + TiO_2	5	Liquid	997.5	4175	0.611	0.00114	-	-

transport. In addition, the Water + TiO_2 nanofluid exhibited a high specific heat capacity. It enhanced thermal conductivity, thereby improving convective cooling performance and ensuring more stable thermal regulation throughout the PV cooling process.

The thermal behavior of the nano-enhanced bio-PCM was governed by an energy conservation mechanism that accounted for both sensible and latent heat transfer during phase transitions. The total thermal energy absorbed by the PCM can be expressed using the following Eq. (1) [45,46]:

$$Q = mC_p\Delta T + mL \quad (1)$$

where Q represents the total absorbed thermal energy, mis the PCM mass, C_p is the specific heat capacity, ΔT is the temperature difference, and Ldenotes latent heat during the phase change process. Furthermore, the enhancement of nanofluid thermal conductivity due to nanoparticle dispersion can be estimated using the effective thermal conductivity relationship in Eq. (2) [47,48]:

$$k_{eff} = k_f\left(\frac{k_p + 2k_f - 2\phi(k_f - k_p)}{k_p + 2k_f + \phi(k_f - k_p)}\right) \quad (2)$$

where k_{eff} is the effective thermal conductivity of the nanofluid, k_f is the thermal conductivity of the base fluid, k_p is the thermal conductivity of nanoparticles, and ϕrepresents nanoparticle volume fraction. The incorporation of TiO_2 nanoparticles therefore improved both conductive and convective heat transfer mechanisms within the cooling system. Consequently, the hybrid integration of a nano-enhanced bio-PCM with a TiO_2/water nanofluid provided improved thermal storage capability, enhanced heat dissipation, and more stable PV operating temperatures under fluctuating solar irradiation.

2.4. Boundary conditions and numerical simulation configuration

The thermal boundary conditions and numerical simulation configuration were carefully established to accurately represent the hybrid PV cooling system under realistic operating conditions. The overall CFD computational domain and thermal boundary implementation are illustrated in Fig. 3. In this study, solar irradiation intensities of 600, 800, and 1000 W/m² were applied as surface heat fluxes on the upper PV surface to simulate varying solar exposure conditions during daytime operation. The applied heat flux represented the absorbed solar thermal

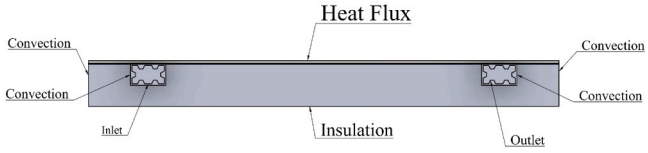


Fig. 3. Thermal boundary conditions and numerical simulation configuration of the hybrid PV cooling system, including heat flux, convection, insulation, inlet, and outlet conditions.

energy contributing directly to PV temperature rise and thermal accumulation within the multilayer structure. Meanwhile, convection boundary conditions with a convection coefficient of 6 W/m²·K and an ambient temperature of 300 K were imposed on the external surfaces of the PV module, the collector region, and the PCM container to represent natural heat dissipation from the system to the surrounding environment. The convective mechanism facilitated heat transfer between the external surfaces and the ambient air, thereby influencing the cooling rate and overall thermal stability of the integrated PV system. Furthermore, an insulating boundary condition was applied to the bottom surface of the system to minimize heat loss to the supporting structure and ensure that most of the thermal energy remained within the proposed cooling configuration.

In this study, the outlet temperature behavior of the nanofluid flow was assumed to be primarily influenced by variations in solar irradiance, as the fluid flow within the collector channel was maintained at steady state. Under steady-flow conditions with a constant mass flow rate of 0.039 kg/s, the thermal response of the outlet region remained relatively stable after the initial transient heating stage. The numerical observations indicated that significant outlet temperature fluctuations occurred mainly during the early thermal accumulation period due to rapid absorption of solar heat and initial temperature adaptation among the PV layer, collector wall, nanofluid region, and nano-enhanced bio-PCM layer. However, after reaching quasi-steady thermal equilibrium, the outlet temperature variation became relatively small and gradually stabilized. Therefore, the outlet temperature distribution in the present simulation was considered to be predominantly dependent on the applied solar irradiation levels rather than on transient hydrodynamic fluctuations. This assumption enabled the thermal prediction framework to focus more effectively on irradiation-induced thermal behavior and

heat dissipation performance of the integrated cooling system, while maintaining computational stability and numerical consistency throughout the simulation.

The phase-change process of the nano-enhanced bio-PCM was modeled using the enthalpy-porosity approach, with the solidus and liquidus temperatures and latent heat values for each PCM composition defined according to the thermophysical properties reported in Table 3. The TiO_2 concentration was set to 5% in both the nano-enhanced bio-PCM and the TiO_2/water nanofluid. The nanofluid was treated as a single-phase homogeneous mixture, while its effective thermophysical properties were defined based on the adopted material configuration. The nanofluid flow was maintained at a constant mass flow rate of 0.039 kg/s under steady hydrodynamic conditions. At the same time, pressure drop and pumping power contributions were not included in the present formulation of thermal performance. They were therefore treated as outside the scope of the current numerical analysis.

The convective heat transfer mechanism applied in this study can be expressed using Newton's law of cooling as follows: Eq. (3) [49,50]:

$$q = hA(T_s - T_\infty) \tag{3}$$

where q represents convective heat transfer rate, h is the convection heat transfer coefficient, Adenotes the heat transfer surface area, T_s is the surface temperature, and T_∞ represents ambient temperature. Meanwhile, the applied solar heat flux boundary condition can be represented by Eq. (4) [29,51]:

$$q'' = \frac{Q}{A} \tag{4}$$

where q'' is the solar heat flux applied to the PV surface, Q represents absorbed thermal energy, and A is the exposed PV area. The insulation boundary principle assumes negligible heat transfer across the insulated surface, which can be mathematically expressed as Eq. (5) [52,53]:

$$\frac{\partial T}{\partial n} = 0 \tag{5}$$

where $\frac{\partial T}{\partial n}$ represents the temperature gradient normal to the insulated surface. This assumption ensured that thermal energy remained confined within the computational domain and improved the accuracy of the evaluation of the proposed cooling mechanism.

The numerical simulation was conducted using ANSYS Fluent with a transient, three-dimensional CFD approach to analyze the thermal response of the integrated PV cooling system over 180 min. The computational mesh consisted of 129955 nodes and 422987 elements, with a fine relevance center and a high smoothing configuration to improve mesh quality and numerical precision. A uniform-sized function with an element size of 7.0×10^{-3}m was applied to maintain stable spatial discretization throughout the geometry domain. The SIMPLE algorithm was employed for pressure–velocity coupling. At the same time, PRESTO discretization was implemented for the pressure Equations, and second-order upwind schemes were applied to the momentum and energy Equations to improve computational stability and solution accuracy. In addition, convergence criteria were specified as $10^{-6}$for energy Equations and $10^{-5}$for continuity and momentum, equations to ensure numerical consistency during the iterative solution process. Under-relaxation factors of 0.3, 0.5, and 0.9 were assigned to the pressure, momentum, and energy Equations, respectively, to maintain numerical stability and accelerate convergence during transient thermal simulations.

2.5. Thermal and electrical performance evaluation

The thermal and electrical performance of the proposed hybrid PV cooling system was evaluated using formulations that explicitly account for the temperature-dependent electrical response and the total useful thermal energy absorbed by both the nano-enhanced bio-PCM and the TiO_2/water nanofluid. The previous thermal-efficiency formulation considered only the PCM's sensible-heat contribution and therefore did not account for the latent heat absorbed during the phase transition or the useful heat carried away by the circulating nanofluid. To provide a physically consistent evaluation of the hybrid cooling architecture, the thermal-energy balance was revised by incorporating both sensible and latent heat storage in the PCM and the sensible heat gained by the nanofluid. This formulation enables the calculated thermal efficiency to represent the total useful thermal energy recovered from the PV cooling system, rather than only the PCM's sensible heating. The electrical efficiency was evaluated independently because PV electrical conversion is primarily affected by the PV module's operating temperature. Accordingly, the combined thermal and electrical evaluation provides a more complete representation of the energy-conversion performance of the proposed hybrid cooling configuration under different solar irradiation conditions.

The electrical efficiency of the PV system was calculated using the temperature-dependent relationship given in Eq. (6) [54,55]:

$$\eta_{el=} \quad \eta_{ref}\left[1 - \beta_{ref}\left(T_o - T_{ref}\right)\right] \tag{6}$$

where η_{el} represents the PV electrical efficiency, η_{ref} is the reference electrical efficiency with a value of 0.2, β_{ref} is the temperature coefficient equal to 0.004, T_o represents the PV operating temperature, and T_{ref} is the reference temperature of 298.15 K [54–56]. This relationship describes the reduction in electrical efficiency as PV operating temperature increases above the reference condition. Therefore, effective thermal regulation that suppresses excessive PV temperature rise can reduce temperature-induced electrical losses. The use of this formulation enables consistent quantification of the electrical performance of the proposed cooling configuration across different irradiation conditions. The electrical efficiency calculation was performed using the PV operating temperature obtained from the transient thermal analysis. The resulting electrical performance was subsequently evaluated together with the revised thermal-efficiency formulation to characterize the overall energy-conversion behavior of the hybrid PV cooling system.

For thermal performance evaluation, the total useful thermal energy was calculated by accounting for the sensible and latent heat absorbed by the nano-enhanced bio-PCM, as well as the useful sensible heat transferred to the TiO_2/water nanofluid. The thermal energy absorbed by the nano-enhanced bio-PCM was expressed using an enthalpy-based formulation that accounts for both temperature-dependent sensible heat and latent heat during phase transition. The PCM thermal-energy contribution can therefore be expressed as Eq. (7):

$$Q_{PCM} m_{PCM}\left[c_{p,PCM}\left(T_o - T_{ref}\right) + f_m L_{PCM}\right] \tag{7}$$

where Q_{PCM} represents the total thermal energy absorbed by the nano-enhanced bio-PCM, m_{PCM} is the PCM mass, $c_{p,PCM}$is the specific heat capacity of the PCM, T_o is the PV operating temperature, T_{ref} is the reference temperature, f_m is the melting fraction of the PCM, and L_{PCM} is the latent heat of fusion. The first term represents the sensible heat absorbed as the PCM temperature increases, whereas the second term represents the latent heat absorbed during the phase-transition process. This formulation is particularly important for the present hybrid system because the transient cooling process involves partial and progressive PCM melting under continuous solar irradiation. Therefore, excluding the latent heat contribution would underestimate the actual thermal energy absorbed by the PCM layer. The use of the melting fraction also ensures that the thermal-energy calculation remains consistent with the transient phase-change behavior obtained from the CFD simulation.

The useful thermal energy transferred to the circulating TiO_2/water nanofluid was calculated separately from the PCM contribution using the sensible heat gained by the nanofluid in Eq. (8):

$$Q_{nf} m_{nf} c_{p,nf}\left(T_{out} - T_{in}\right) \tag{8}$$

where Q_{nf} represents the useful thermal energy carried by the nanofluid, m_{nf} is the nanofluid mass circulated during the evaluation period, $c_{p,nf}$ is the specific heat capacity of the TiO_2/water nanofluid, T_{out} is the nanofluid outlet temperature, and T_{in} is the nanofluid inlet temperature. The total useful thermal energy of the hybrid cooling system was then obtained by combining the thermal contributions of the nano-enhanced bio-PCM and TiO_2/water nanofluid. Accordingly, the revised thermal efficiency was defined as Eq. (9):

$$\eta_{th}\frac{Q_{PCM}+Q_{nf}}{IAt} \tag{9}$$

where η_{th} represents the total thermal efficiency, I is the solar irradiation intensity, A is the PV surface area, and t is the evaluation time. This formulation explicitly includes both latent and sensible heat stored in the PCM and the useful heat transported by the nanofluid, thereby providing a more complete energy balance for the hybrid cooling system. The denominator IAt represents the total incident solar energy over the PV surface during the corresponding evaluation period. Higher η_{th} Therefore, it indicates that a larger fraction of the incident solar energy is captured as useful thermal energy through the combined PCM storage and nanofluid heat-removal mechanisms [57,58]. This revised formulation was adopted for the thermal performance evaluation to ensure that the reported thermal efficiency reflects the full thermal contribution of both cooling components, rather than the PCM's sensible-heat contribution alone.

2.6. Hybrid RNN–LSTM architecture for thermal–electrical prediction

The hybrid RNN–LSTM architecture was developed to predict the thermal–electrical behavior of the proposed hybrid PV cooling system using CFD-generated simulation data. The overall prediction framework is illustrated in Fig. 4, consisting of sequential feature extraction with RNN layers, temporal sequence learning with LSTM layers, and the generation of the final prediction. In this study, the input dataset was constructed using several important thermal–electrical parameters obtained from transient CFD simulations, including PV temperature, PCM melting behavior, outlet temperature, electrical efficiency, and thermal efficiency under different solar irradiation conditions. The input sequences were generated using sliding temporal windows to capture dynamic thermal variations and nonlinear operating behavior of the PV cooling system. Furthermore, solar irradiation levels of 600, 800, and 1000 W/m^2 were included in the dataset to improve the model's learning under varying environmental conditions. This hybrid deep learning framework was designed to improve prediction accuracy, temporal learning stability, and adaptive forecasting capability for intelligent PV thermal management applications. However, because the CFD dataset was generated from a deterministic factorial design without replicated simulation runs, it lacks the independent experimental replication needed to estimate pure error. Therefore, the prediction framework should be interpreted as learning deterministic CFD-generated temporal patterns rather than statistically independent replicated observations, and potential unmodelled interaction effects should be recognized as a limitation of the underlying dataset.

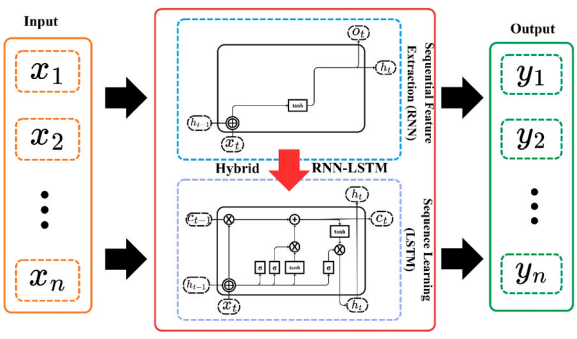


Fig. 4. Hybrid RNN–LSTM architecture for thermal–electrical prediction.

The dataset was constructed using sequential time-series features extracted from CFD simulation results. In the implemented framework, the previous five PV temperature values and the previous five outlet temperature values were used as temporal input features, along with the irradiation intensity, electrical efficiency, and thermal efficiency parameters. The model's target output was the future PV temperature prediction at the next time step. Before training, all datasets were normalized using MinMaxScaler to improve numerical stability and accelerate convergence. The normalized data can be expressed as follows: Eq. (10) [59,60]:

$$X_{norm}=\frac{X-X_{min}}{X_{max}-X_{min}} \tag{10}$$

where X_{norm} represents normalized data, X is the original input value, while X_{min} and X_{max}respectively denote the minimum and maximum values in the dataset. This normalization process ensured that all input parameters possessed comparable numerical ranges, thereby improving model learning performance and reducing instability during gradient optimization.

The sequential prediction mechanism in the hybrid RNN–LSTM model was designed to capture temporal dependencies and nonlinear thermal behavior from CFD-generated datasets. In the RNN stage, sequential information from previous thermal states was processed recursively to extract short-term temporal patterns. The hidden state update mechanism of the RNN layer can be expressed as Eq. (11) [61,62]:

$$h_t=f(W_{xh}x_t+W_{hh}h_{t-1}+b_h) \tag{11}$$

where $h_t$represents the hidden state at time step t, $x_t$denotes input data, $W_{xh}$and $W_{hh}$are weight matrices, and $b_h$represents bias parameters. The activation function futilized the hyperbolic tangent (tanh) function to improve nonlinear learning capability. After sequential feature extraction by the RNN layer, the processed temporal features were transferred into the LSTM stage to learn long-term dependencies and thermal evolution characteristics more effectively.

The LSTM component preserved important temporal information while minimizing gradient vanishing, a problem commonly encountered in conventional RNN architectures. The forget gate mechanism within the LSTM structure can be expressed as follows: Eq. (12) [63,64]:

$$f_t=\sigma(W_f[h_{t-1},x_t]+b_f) \tag{12}$$

while the cell state updating process can be represented by Eq. (13) [65,66]:

$$C_t=f_t\odot C_{t-1}+i_t\odot\widetilde{C}_t \tag{13}$$

where f_t represents the forget gate, C_t denotes cell memory state, i_t is the input gate, and $\widetilde{C}_t$ represents candidate memory information. Through this mechanism, the hybrid RNN–LSTM architecture learned both short-term thermal fluctuations and long-term thermal evolution patterns simultaneously. The implemented deep learning model used three hidden layers with 64, 48, and 32 neurons, tanh activation functions, and the Adam optimization algorithm. The training process was performed for 100 epochs, with an adaptive learning rate initialized at 0.001 to improve convergence and computational stability.

The prediction output generated by the hybrid RNN–LSTM model was subsequently evaluated using several statistical performance indicators, including MSE, RMSE, MAE, and R^2. The MSE formulation used in this study can be expressed as follows: Eq. (14) [67,68]:

$$MSE=\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat{y}_i)^2 \tag{14}$$

while RMSE was calculated using Eq. (15) [67,68]:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{15}$$

where y_i represents actual temperature values and $\hat{y}_i$ denotes predicted temperature values. These evaluation metrics were used to quantify prediction accuracy and to assess the proposed hybrid deep learning framework's capability to forecast PV thermal behavior under varying environmental conditions. Because electrical and thermal efficiency are derived from the same CFD thermal state and may contain information associated with the target PV temperature, including them as input features can introduce target-related information and potentially inflate the apparent predictive performance. In addition, the use of adjacent sliding temporal windows requires strict separation of sequential samples between training and testing partitions to prevent overlapping temporal information from appearing across partitions and causing data leakage. Therefore, integrating CFD simulation with a hybrid RNN–LSTM prediction architecture provided an intelligent, adaptive framework to improve the reliability and operational stability of next-generation PV cooling systems.

To establish a physically traceable basis for the annual sustainability predictions, the predicted annual electricity savings were determined by integrating the instantaneous electrical energy benefit over the annual operating period, while the corresponding CO_2 reduction was calculated from the avoided electricity consumption using an emission factor. The annual electricity saving can be expressed as Eq. (16), and the associated CO_2 reduction can be estimated using Eq. (17), where E_{save} represents annual electricity saving, $P_{save}(t)$ denotes the instantaneous power-saving benefit, EF_{CO_2} represents the applicable electricity emission factor, and $CO_{2,save}$ denotes the avoided CO_2 emissions. These relationships provide an explicit, physical accounting pathway that links the predicted thermal–electrical improvement to annual energy and environmental benefits, rather than treating the sustainability indicators as independent prediction outputs.

$$E_{save} = \int_0^T P_{save}(t)\,dt \tag{16}$$

$$CO_{2,save} = E_{save} \times EF_{CO_2} \tag{17}$$

2.7. Validation and mesh independence analysis

The validation analysis evaluated the numerical accuracy and reliability of the proposed CFD simulation framework across different thermal management configurations. The comparison between reference simulation data and the current numerical simulation results for several PCM materials is shown in Fig. 5**(a)**. The validation process demonstrated that the obtained temperature deviations remained very small, with error values ranging from 1.90% to 1.99% for the RT-42, $Na_2HPO_4{\cdot}12H_2O$, and $NaNO_3$–NaCl PCM configurations. These results indicate strong agreement between the developed numerical model and previously reported reference simulations. The low error magnitude confirmed that the implemented thermal boundary conditions, material properties, and numerical discretization schemes accurately represented the thermal behavior of the integrated PV cooling system. Furthermore, the MAPE values obtained remained significantly below the acceptable validation threshold of 10%, demonstrating that the proposed simulation framework exhibits high predictive capability and strong numerical consistency for thermal performance evaluation. Therefore, the validation results strongly support the reliability and credibility of the developed CFD model for analyzing nano-enhanced PV cooling systems under varying operating conditions.

Additional validation of nanofluid thermal performance across different nanoparticle volume fractions was conducted to further evaluate the robustness of the numerical model. As illustrated in Fig. 5(b), the comparison between the reference and validation simulations for nanoparticle concentrations of 0.6%, 1%, 1.6%, and 2% yielded errors ranging from 1.62% to 1.75%. The small deviation between the datasets indicates that the implemented numerical approach accurately captured the thermal enhancement effects caused by nanoparticle dispersion in the cooling fluid. In addition, the stable error distribution across different nanoparticle concentrations confirmed that the thermophysical property implementation and the convective heat transfer modeling remained numerically stable throughout the simulation. The obtained validation results also demonstrated that the proposed CFD framework can effectively reproduce the thermal response behavior of nanofluid-based cooling systems with high computational accuracy. Consequently, the low MAPE values obtained in this study further strengthen the confidence in the developed hybrid PV thermal simulation model and support its applicability for advanced thermal management investigations.

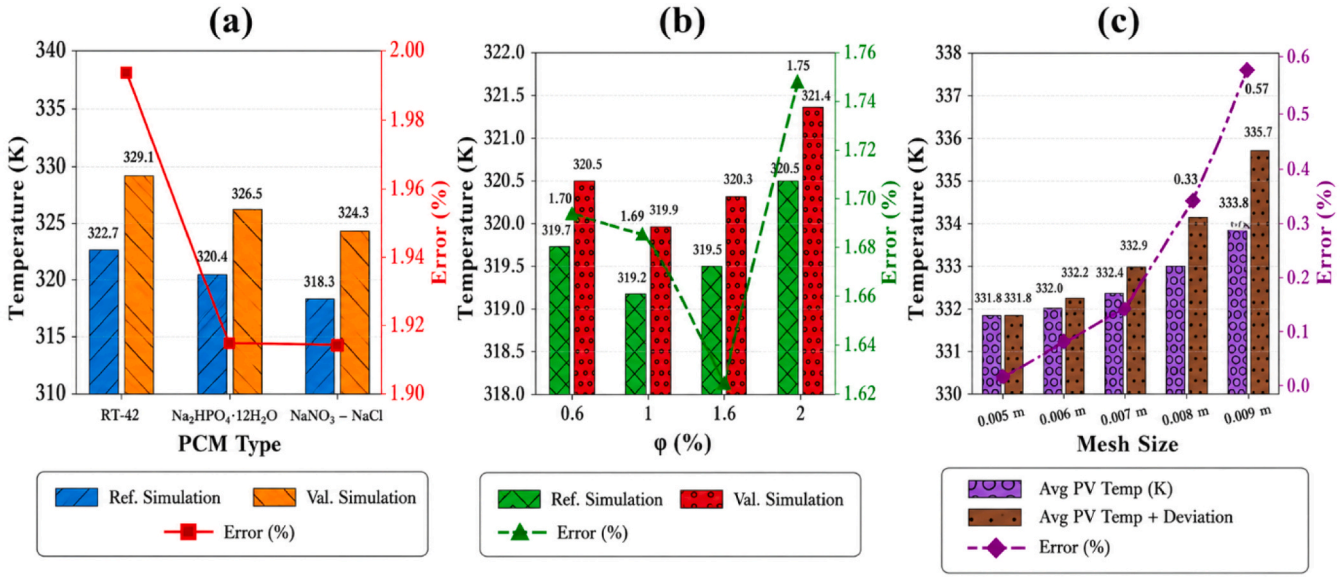


Fig. 5. Validation and mesh independence analysis of the hybrid PV cooling system: (a) Validation of PCM thermal simulation results [69], (b) Validation of nanofluid thermal performance under different nanoparticle volume fractions [70], and (c) Mesh independence analysis based on average PV temperature deviation.

A mesh independence study was subsequently conducted to determine the optimal mesh configuration that provides accurate numerical results while maintaining computational efficiency. The mesh sensitivity analysis shown in Fig. 5(c) evaluated several mesh sizes ranging from 0.005 m to 0.009 m based on the average PV temperature deviation characteristics. Although the mesh size of 0.005 m achieved the highest numerical accuracy, the computational cost was considerably higher due to the increased element density and longer simulation time. Meanwhile, larger mesh sizes above 0.007 m resulted in progressively larger temperature deviations and reduced numerical precision, due to insufficient spatial resolution to capture thermal gradients within the multilayer cooling structure. Therefore, the mesh size of 0.007 m was selected as the optimal computational configuration, providing an excellent balance between numerical accuracy and computational efficiency. At this mesh size, the temperature deviation remained very small at 0.5469 K, corresponding to an error of 0.16%, indicating that further mesh refinement would yield only a negligible improvement in numerical accuracy while substantially increasing the computational burden. Hence, the selected mesh configuration ensured stable convergence, reliable thermal predictions, and efficient computational performance throughout the transient CFD simulation.

3. Results

3.1. Effect of solar irradiation and nano-enhanced bio-PCM on PV temperature distribution

The transient thermal behavior of the hybrid PV cooling system under a solar irradiance of 600 W/m^2 is presented in Fig. 6**(a)**. The results demonstrate that all nano-enhanced bio-PCM configurations experienced gradual temperature increases over the 180-minute simulation period due to the continuous accumulation of heat from absorbed solar energy. However, significant differences in thermal regulation capability were observed among the investigated materials. Coconut Oil-TiO_2 consistently exhibited the lowest PV operating temperature compared to Palmitic Acid-TiO_2, Lauric Acid-TiO_2, and Beeswax-TiO_2 during the entire transient process. At the end of the simulation, Coconut Oil-TiO_2 achieved a final PV temperature of approximately 313.07 K, whereas Lauric Acid-TiO_2 and Beeswax-TiO_2 reached higher temperatures of approximately 316.95 K and 316.48 K, respectively. The lower temperature observed in Coconut Oil-TiO_2 indicates superior thermal absorption and heat dissipation, resulting from more stable latent heat storage behavior and improved thermal diffusion. In addition, the relatively slower temperature increase in Coconut Oil-TiO_2 confirms that integrating a nano-enhanced bio-PCM with nanofluid cooling successfully reduced excessive thermal accumulation within the PV structure. This behavior can be attributed to the PCM layer's ability to absorb part of the accumulated thermal energy as latent heat during the phase transition. At the same time, the TiO_2/water nanofluid simultaneously facilitates convective heat removal from the PV structure. The lower temperature profile of Coconut Oil-TiO_2 therefore indicates that the combined latent heat absorption and nanofluid-assisted heat transfer were more effective at maintaining thermal stability than in the other investigated configurations. These findings indicate that the thermophysical properties of the selected bio-PCM strongly influence transient thermal stabilization and directly affect PV cooling performance under moderate irradiation conditions.

The thermal response characteristics under intermediate irradiation conditions at 800 W/m^2 are illustrated in Fig. 6**(b)**. Overall, the increase in solar irradiance led to higher PV operating temperatures across all investigated cooling configurations, due to greater thermal energy absorbed by the PV surface. Nevertheless, the hybrid cooling architecture continued to demonstrate effective thermal regulation throughout the transient simulation. Similar to the previous condition, Coconut Oil-TiO_2 maintained the lowest temperature profile among all nano-enhanced bio-PCM materials, indicating stable thermal management performance even under increased thermal loading conditions. Meanwhile, Lauric Acid-TiO_2 exhibited the largest temperature rise, suggesting lower thermal buffering capacity and reduced effectiveness in suppressing thermal accumulation compared with the other materials investigated. The temperature differences observed between the bio-PCM materials became increasingly pronounced after the initial heating stage, as the latent heat storage mechanisms and melting characteristics of each material began to dominate the thermal regulation process. Furthermore, the temperature distribution indicates that the combined interaction between latent heat absorption from the bio-PCM and convective cooling from the TiO_2/water nanofluid successfully delayed rapid thermal escalation within the PV module. The increasingly distinct temperature profiles after the initial heating period suggest that PCM thermal response differences become more influential as

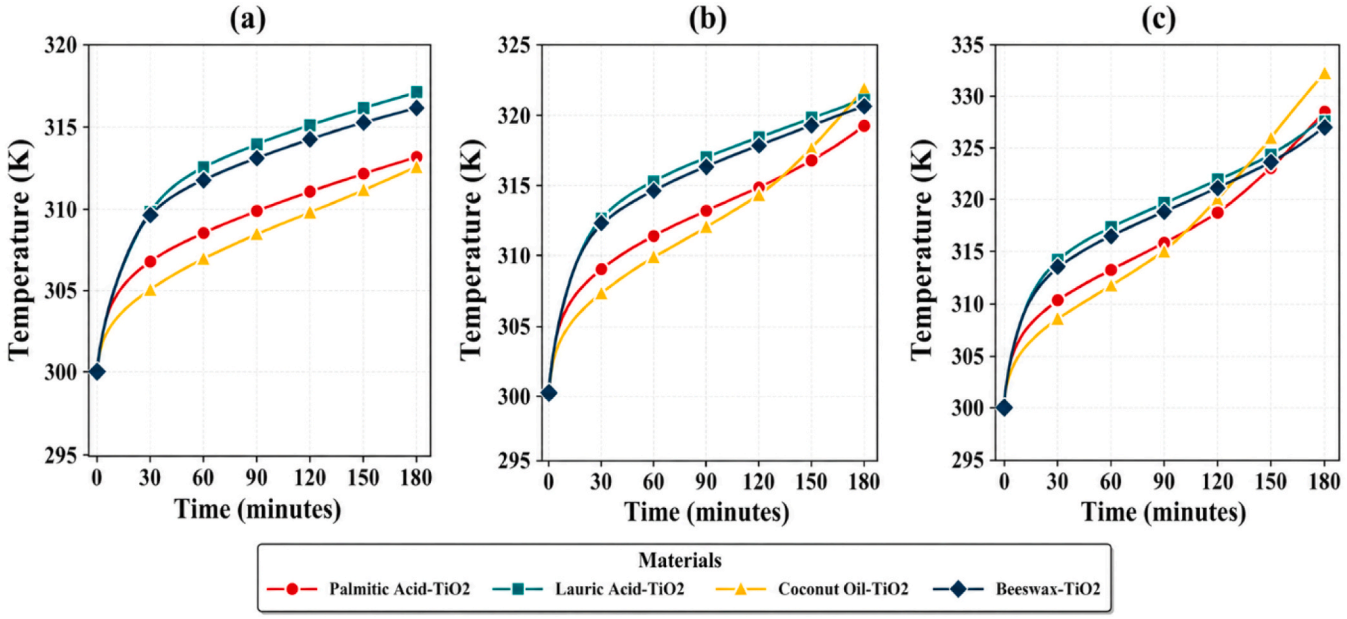


Fig. 6. Transient PV temperature distribution under different solar irradiation conditions using nano-enhanced bio-PCM materials: (a) 600 W/m^2, (b) 800 W/m^2, and (c) 1000 W/m^2.

the accumulated heat approaches the phase-transition range. Under this condition, a more effective latent heat absorption process can temporarily limit the increase in PV temperature, while the nanofluid maintains continuous heat transport away from the heated region. Consequently, the proposed hybrid cooling system demonstrated strong thermal stability and maintained relatively controlled temperature evolution despite increased irradiation intensity.

The highest irradiation condition, 1000 W/m^2, produced the most significant thermal loading behavior, as shown in Fig. 6**(c)**. Under these conditions, all PV configurations experienced considerably larger temperature increases due to intensified solar heat flux and prolonged thermal accumulation over the 180-minute simulation period. Beeswax-TiO_2 demonstrated the thermal regulation performance associated with the reported final PV temperature of approximately 327.29 K. At the same time, Palmitic Acid-TiO_2, Lauric Acid-TiO_2, and Coconut Oil-TiO_2 exhibited final PV temperatures of approximately 328.68 K, 328.12 K, and 332.39 K, respectively. These results indicate that Beeswax-TiO_2 maintained the lowest final PV temperature among the reported configurations under the highest irradiation condition, suggesting that its thermal storage characteristics and TiO_2-enhanced heat transfer remained effective in limiting temperature escalation under severe thermal loading. In contrast, the higher final temperature of Coconut Oil-TiO_2 at 332.39 K indicates that its transient thermal regulation performance was comparatively less effective at 1000 W/m^2, despite its favorable behavior under other operating conditions. At 1000 W/m^2, the substantially higher thermal input underscores the importance of both latent heat utilization and heat-transfer efficiency, as the cooling system must continuously accommodate a larger thermal load. The relatively low temperature maintained by Beeswax-TiO_2 indicates that its latent heat storage capability, combined with TiO_2-enhanced thermal transport, effectively absorbed and redistributed heat during the high-irradiation period. Moreover, the temperature response demonstrates that the relative effectiveness of each nano-enhanced bio-PCM can vary with irradiation intensity, emphasizing the importance of evaluating cooling performance across multiple operating conditions rather than relying on a single irradiation level. Therefore, the results demonstrate that the proposed nano-enhanced bio-PCM and nanofluid cooling framework can improve PV thermal stability and limit excessive operating temperatures under varying solar irradiation environments, with Beeswax-TiO_2 exhibiting the lowest reported final PV temperature at 1000 W/m^2.

The thermal contour distribution in Fig. 7(a) shows that the PV temperature field at 600 W/m^2 remained relatively uniform across all nano-enhanced bio-PCM configurations, indicating stable heat absorption during the initial irradiation stage. The contour profiles of Palmitic Acid-TiO_2 and Lauric Acid-TiO_2 showed moderate temperature propagation, with dominant green-cyan regions, suggesting gradual heat accumulation across the PV surface. Coconut Oil-TiO_2 showed a more localized contour distribution with narrower high-temperature regions, indicating lower thermal diffusion than the other nano-enhanced bio-PCM combinations. In contrast, Beeswax-TiO_2 produced a broader temperature contour distribution and a slightly higher thermal intensity, consistent with previous temperature data showing that Beeswax-TiO_2 achieved the highest PV temperature under identical irradiation conditions. The contour homogeneity observed under this irradiation condition confirms that the addition of TiO_2 nanoparticles enhanced thermal conductivity and improved heat-spreading behavior within the nano-enhanced bio-PCM matrix. These findings are strongly correlated with the previously discussed average PV temperature trend, in which the temperature increase at 600 W/m^2 occurred progressively and

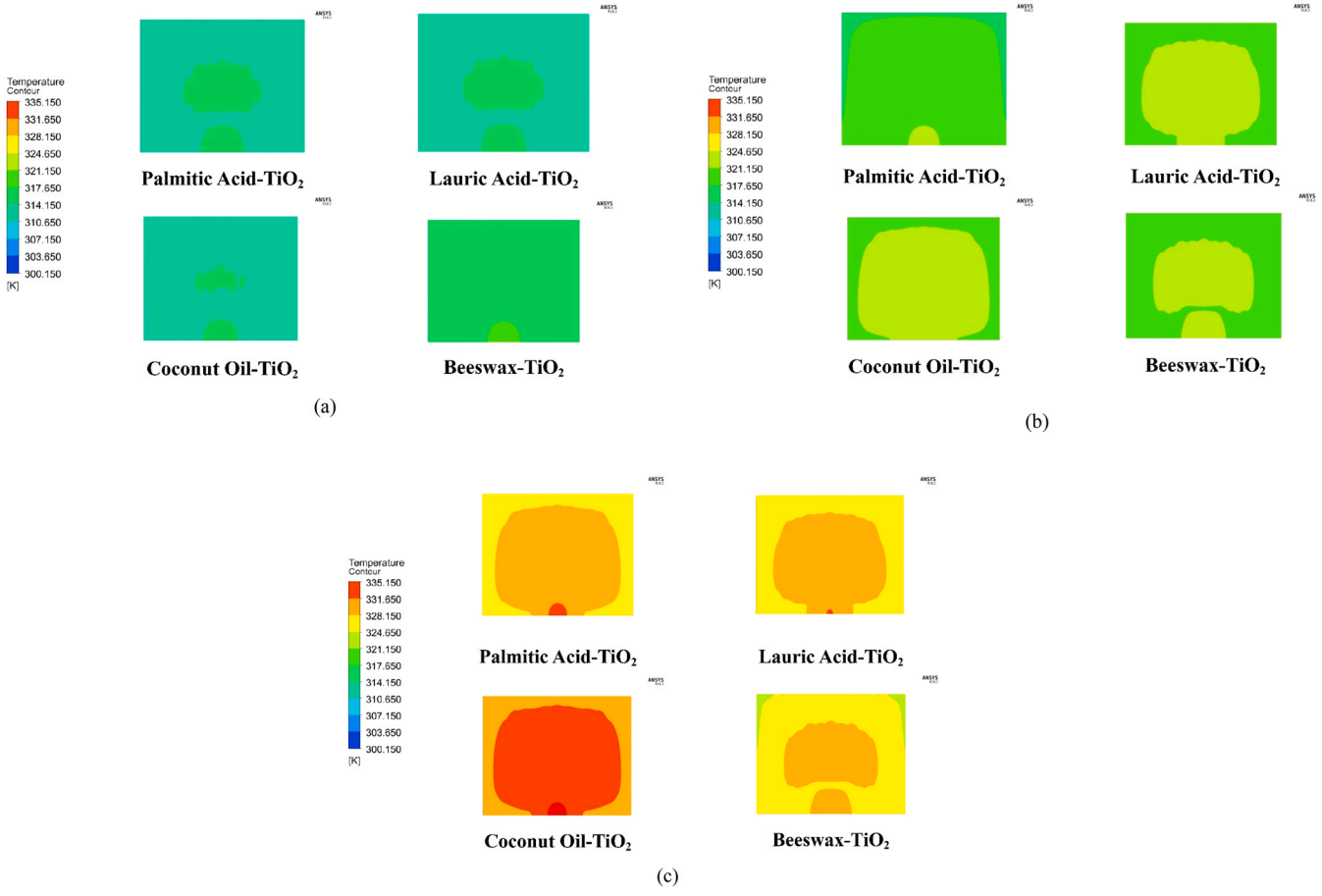


Fig. 7. Temperature contour distribution of nano-enhanced bio-PCM at solar irradiation intensities of (a) 600 W/m^2, (b) 800 W/m^2, and (c) 1000 W/m^2.

remained thermally stable throughout the simulation.

The contour evolution shown in Fig. 7**(b) and** Fig. 7**(c)** indicates a substantial increase in thermal intensity as solar irradiation increased from 800 W/m^2 to 1000 W/m^2. At 800 W/m^2, the contour regions transitioned from green-dominant distributions into yellow-green patterns, reflecting intensified heat accumulation and wider thermal propagation across the PV module surface. Coconut Oil-TiO_2 exhibited a larger high-temperature region at this stage, indicating improved thermal absorption at medium irradiation intensity, whereas Beeswax-TiO_2 maintained a balanced contour distribution and stable thermal spreading characteristics. Furthermore, at 1000 W/m^2 in Fig. 7(c), all nano-enhanced bio-PCM configurations exhibited significant contour expansion toward orange and red regions, confirming rapid thermal accumulation driven by elevated solar flux. Coconut Oil-TiO_2 produced the highest contour intensity, with dominant orange-red regions, which aligns with previous quantitative data indicating the highest final PV temperature among all samples at 1000 W/m^2. Meanwhile, Lauric Acid-TiO_2 and Palmitic Acid-TiO_2 displayed lower contour intensities, indicating more controlled thermal distribution despite higher irradiation levels. Overall, the contour analysis confirms that increasing irradiation intensity directly intensified PV thermal accumulation. In contrast, the thermal behavior of each nano-enhanced bio-PCM composite remained strongly dependent on its heat storage capacity, thermal conductivity, and heat dissipation characteristics.

3.2. Effect of solar irradiation on melting fraction characteristics of nano-enhanced bio-PCM

The melting fraction behavior under solar irradiation of 600 W/m^2 demonstrates that the thermal absorption capability of each nano-enhanced bio-PCM strongly influences the transient melting process within the hybrid PV cooling system, as shown in Fig. 8**(a)**. The results indicate that Coconut Oil-TiO_2 exhibited the fastest melting progression throughout the 180-minute simulation period, reaching a melting fraction of approximately 0.762, which was significantly higher than Palmitic Acid-TiO_2, Lauric Acid-TiO_2, and Beeswax-TiO_2. In contrast, Lauric Acid-TiO_2 exhibited the slowest melting response, with a final melting fraction below 0.411, indicating slower latent heat activation and reduced thermal absorption under moderate irradiation conditions. The relatively rapid melting evolution observed in Coconut Oil-TiO_2 confirms that the material possessed superior thermal diffusivity and more effective latent heat utilization during continuous heat exposure. Furthermore, the gradual melting of Beeswax-TiO_2 suggests that although the material was capable of storing thermal energy, its phase transition occurred more slowly due to lower thermal transport within the PCM matrix. This difference in melting progression is primarily due to each PCM configuration's ability to transfer absorbed heat from the PV interface to the PCM region, thereby determining how rapidly the latent heat storage mechanism is activated. The higher melting fraction of Coconut Oil-TiO_2 indicates that heat was distributed more effectively throughout the PCM domain, allowing a larger portion of the material to undergo phase transition during the same exposure period. The overall melting trends also reveal that the addition of TiO_2 nanoparticles enhanced internal heat diffusion, allowing the melting front to propagate more uniformly throughout the nano-enhanced bio-PCM layer. These observations are consistent with previously observed PV thermal distribution behavior, in which improved thermal absorption characteristics led to more stable transient temperature regulation during prolonged irradiation.

At intermediate irradiation intensity, the evolution of the melting fraction accelerated considerably for all investigated nano-enhanced bio-PCM materials, as illustrated in Fig. 8**(b)**. The increase in absorbed solar energy intensified the latent-heat activation process, leading to faster propagation of melting across the PCM domain during the transient period. Coconut Oil-TiO_2 again demonstrated the highest melting performance, achieving a final melting fraction of approximately 0.933 at 180 min, indicating highly efficient thermal energy absorption and rapid phase transition characteristics under elevated thermal loading. Meanwhile, Beeswax-TiO_2 and Lauric Acid-TiO_2 exhibited comparatively slower melting rates, reaching approximately 0.649 and 0.649, respectively, suggesting lower thermal responsiveness despite the higher irradiation conditions. Palmitic Acid-TiO_2 exhibited intermediate melting behavior, with a relatively stable progression of phase transitions throughout the simulation. In addition, the melting curves reveal that the differences among the nano-enhanced bio-PCM materials became increasingly significant after the initial heating stage, when latent heat storage mechanisms began to dominate thermal regulation. The stronger melting response at 800 W/m^2 can be explained by the greater thermal input available to overcome the energy required for phase transition. In contrast, the enhanced thermal diffusion provided by TiO_2 facilitates the redistribution of heat within the PCM layer. Consequently, the substantially higher melting fraction of Coconut Oil-TiO_2 indicates that its thermal response remained effective as the heat load increased, enabling more extensive utilization of its latent heat

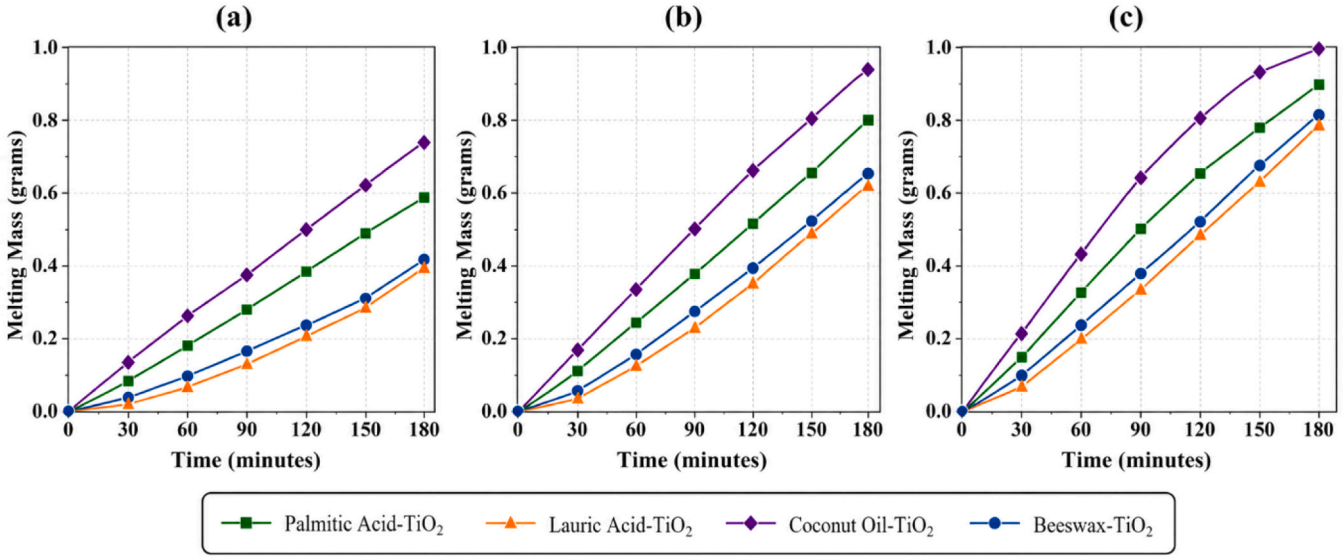


Fig. 8. Transient melting fraction distribution of nano-enhanced bio-PCM under different solar irradiation conditions: (a) 600 W/m^2, (b) 800 W/m^2, and (c) 1000 W/m^2.

capacity. The stable, continuous melting trend observed for Coconut Oil-TiO_2 indicates a strong synergistic interaction between latent heat absorption and nanoparticle-enhanced thermal conductivity, thereby improving heat-transfer uniformity within the cooling structure. Consequently, the melting characteristics under 800 W/m^2 further confirm the important role of thermophysical properties in determining the effectiveness of nano-enhanced bio-PCM for PV thermal management applications.

Under the highest irradiation intensity of 1000 W/m^2, all nano-enhanced bio-PCM configurations experienced substantial melting acceleration due to intensified thermal accumulation and increased solar heat flux, as shown in Fig. 8**(c)**. Coconut Oil-TiO_2 achieved complete melting near the end of the simulation period, with a melting fraction approaching 1.0, demonstrating exceptional latent heat absorption and highly efficient thermal diffusion under severe operating conditions. Palmitic Acid-TiO_2 also exhibited a high melting fraction exceeding 0.948, followed by Lauric Acid-TiO_2 and Beeswax-TiO_2 with final melting fractions of approximately 0.843 and 0.842, respectively. The rapid phase-transition behavior observed at this irradiation level indicates that increasing solar intensity significantly accelerated the heat-storage activation process within the nano-enhanced bio-PCM layer. Moreover, the near-complete melting observed in Coconut Oil-TiO_2 confirms its superior ability to absorb and redistribute the excess thermal energy generated by the PV module during prolonged irradiation. The near-complete melting of Coconut Oil-TiO_2 indicates that the available thermal energy was sufficient to activate most of its latent heat storage capacity. In contrast, the enhanced heat diffusion promoted a more uniform progression of the melting front. In contrast, the lower melting fractions of Lauric Acid-TiO_2 and Beeswax-TiO_2 indicate that a smaller proportion of their PCM domains reached the phase-transition condition within the same simulation period, thereby limiting the extent of latent heat utilization. The comparatively slower melting progression of Beeswax-TiO_2 suggests that, although the material possesses substantial latent heat capacity, its thermal conductivity may limit the overall melting rate under extreme thermal loading conditions. Overall, the melting fraction analysis demonstrates that irradiation intensity strongly affects PCM phase transition dynamics, while integrating TiO_2 nanoparticles successfully improves thermal diffusion and enhances the overall thermal management capability of the hybrid PV cooling architecture. These findings further support the previous thermal contour and temperature distribution analyses, in which materials exhibiting faster, more stable melting behavior also provided improved transient thermal regulation within the PV system.

The mass-fraction contour distribution at a solar irradiance of 600 W/m^2 is shown in Fig. 9(a). The contour results demonstrate that the investigated nano-enhanced bio-PCM materials exhibit distinct phase-transition distributions that depend on their thermophysical characteristics and thermal diffusion capabilities. Coconut Oil-TiO_2 exhibited the most dominant red contour region with mass fraction values approaching 1.0, indicating more extensive melting propagation and stronger latent heat activation throughout the thermal storage region. Meanwhile, Lauric Acid-TiO_2 showed larger green-yellow contour regions, suggesting slower melting development and less uniform thermal diffusion within the cooling structure. Palmitic Acid-TiO_2 and Beeswax-TiO_2 exhibited intermediate contour behavior, with relatively balanced melting propagation throughout the PCM domain. The contour uniformity observed in Coconut Oil-TiO_2 confirms that enhanced thermal conductivity and stable heat-absorption capability promote more effective heat transfer during continuous thermal exposure. These contour characteristics are consistent with the previous transient melting fraction analysis, which showed that Coconut Oil-TiO_2 exhibited the fastest melting progression and superior thermal absorption performance under moderate irradiation conditions.

The contour evolution at irradiation intensities of 800 W/m^2 and

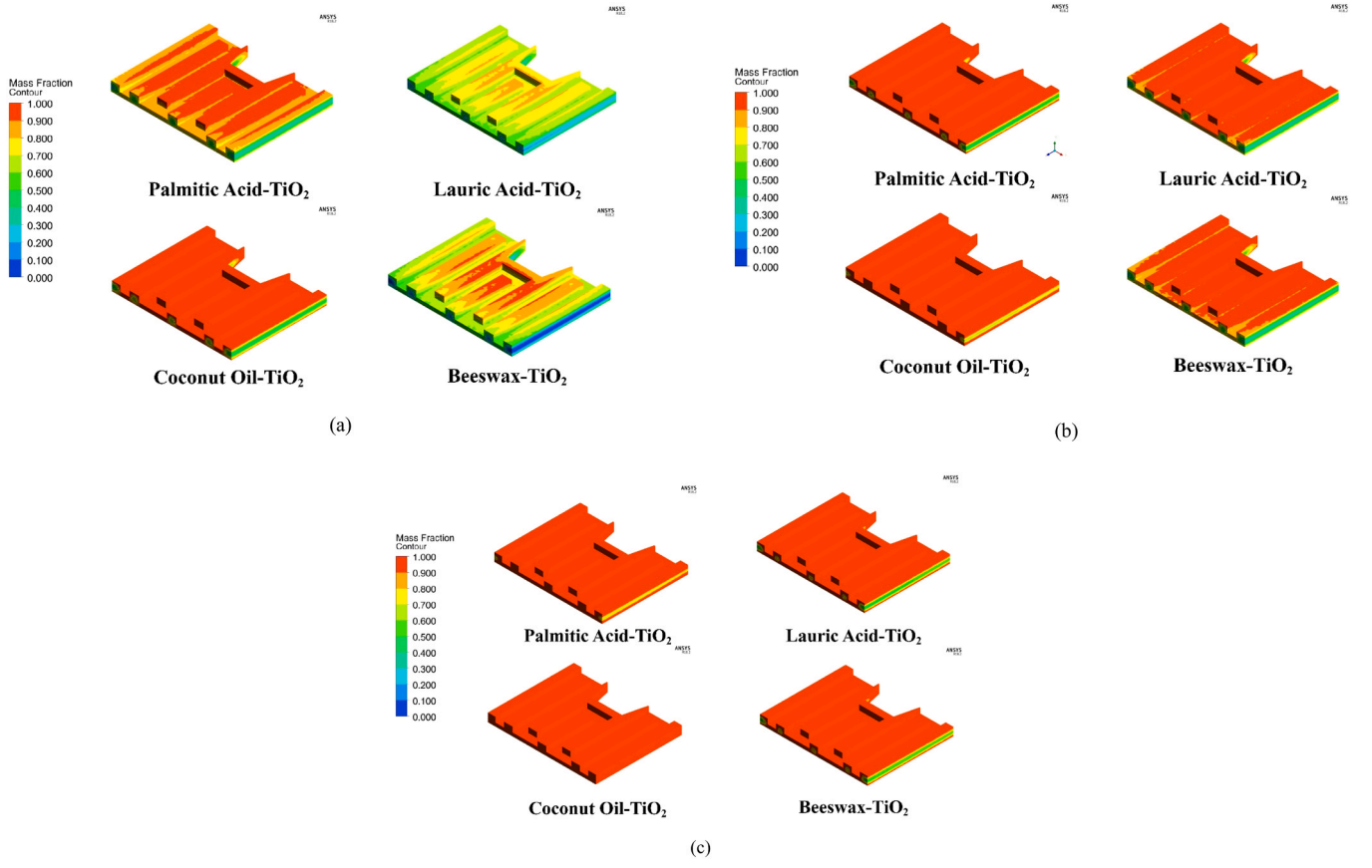


Fig. 9. Mass fraction contour distribution of nano-enhanced bio-PCM under solar Irradiation intensities of (a) 600 W/m^2, (b) 800 W/m^2, and (c) 1000 W/m^2.

1000 W/m² is illustrated in Figs. 9(b) and 9(c), respectively. The increasing solar irradiance substantially altered the mass fraction distribution, leading almost all nano-enhanced bio-PCM configurations to approach fully melted conditions during the transient simulation period. Coconut Oil-TiO_2 maintained the most uniform and dominant high-mass-fraction contour distribution at both irradiation levels, confirming its superior latent heat utilization capability and enhanced thermal diffusion performance under elevated thermal loading. In contrast, Lauric Acid-TiO_2 and Beeswax-TiO_2 still exhibited several localized contour gradients near the cooling boundaries, indicating relatively slower phase transition propagation and less stable heat spreading characteristics compared with Coconut Oil-TiO_2. Furthermore, the contour distributions demonstrate that incorporating TiO_2 nanoparticles significantly accelerated internal thermal diffusion and improved melting uniformity throughout the nano-enhanced bio-PCM layer under severe operating conditions. At 1000 W/m², the contour regions became predominantly red across all investigated materials, confirming rapid thermal accumulation and intensified latent-heat activation due to higher absorbed solar energy. These observations strongly support the previous thermal and melting fraction analyses, which showed that increasing irradiation intensity accelerated the phase transition. In contrast, the thermal regulation capability remained highly dependent on the thermophysical properties of the selected nano-enhanced bio-PCM materials.

3.3. Effect of solar irradiation on nanofluid outlet temperature characteristics

Building on the previously observed PV thermal regulation and melting-fraction behavior of the nano-enhanced bio-PCM layer, the transient outlet-temperature response of the nanofluid was further evaluated under varying solar irradiation intensities. As shown in Fig. 10, the outlet temperature increased rapidly during the initial heating stage, then gradually approached quasi-steady-state conditions over the 180-minute simulation period. At the beginning of the simulation, the outlet temperature remained close to 300 K due to the initial thermal equilibrium among the PV structure, the cooling channel, and the surrounding environment. As the PV surface absorbed thermal energy, heat transfer via conduction and convection gradually increased the outlet temperature. The final outlet temperatures reached approximately 301.41 K, 301.88 K, and 302.35 K under irradiation intensities of 600 W/m², 800 W/m², and 1000 W/m², respectively. Although higher irradiation intensified thermal accumulation, the overall temperature increase remained relatively moderate, indicating stable heat dissipation performance of the integrated cooling system. The decreasing temperature growth rate after the early transient stage also confirms that the thermal interaction among the PV layer, the nano-enhanced bio-PCM, and the circulating nanofluid progressively approached thermal equilibrium.

The outlet temperature behavior was analyzed primarily in terms of solar irradiance, since the hydrodynamic conditions in the collector channel were maintained at a steady state during the numerical simulation. The constant nanofluid mass flow rate of 0.039 kg/s, together with identical convection and insulation boundary conditions, minimized transient flow instability and made irradiation intensity the dominant factor governing thermal accumulation and the evolution of the outlet temperature. The highly similar temperature trends across all irradiation conditions confirm that the external heat flux predominantly controlled the nanofluid's macroscopic thermal response. Furthermore, the relatively small temperature differences indicate that the combined interaction between nano-enhanced bio-PCM latent heat absorption and nanofluid-assisted convective cooling effectively suppressed excessive thermal escalation, demonstrating stable thermal management capability under steady-state operating conditions.

3.4. Effect of solar irradiation on electrical and thermal efficiency of the hybrid PV cooling system

The electrical efficiency characteristics of the hybrid PV cooling system under different solar irradiation intensities are shown in Fig. 11 **(a)**. The results indicate that increasing solar irradiation intensity generally reduced the electrical efficiency of all investigated nano-enhanced bio-PCM configurations due to the continuous rise in PV operating temperature during transient thermal exposure. Under irradiation intensity of 600 W/m², Coconut Oil-TiO_2 achieved the highest electrical efficiency of approximately 18.81%, followed closely by Palmitic Acid-TiO_2 at 18.78%, while Lauric Acid-TiO_2 and Beeswax-TiO_2 exhibited slightly lower efficiencies of 18.50% and 18.53%, respectively. However, as the irradiation intensity increased to 800 W/m² and 1000 W/m², the electrical efficiency gradually decreased for all investigated materials because higher absorbed solar energy intensified thermal accumulation within the PV structure. At the highest irradiation level of 1000 W/m², Beeswax-TiO_2 demonstrated the highest electrical efficiency of approximately 17.67%, whereas Coconut Oil-TiO_2 experienced the lowest value of approximately 17.26%. The non-monotonic variation observed among the individual PCM configurations reflects the competing effects of increased solar energy input and material-dependent thermal regulation, whereby changes in melting

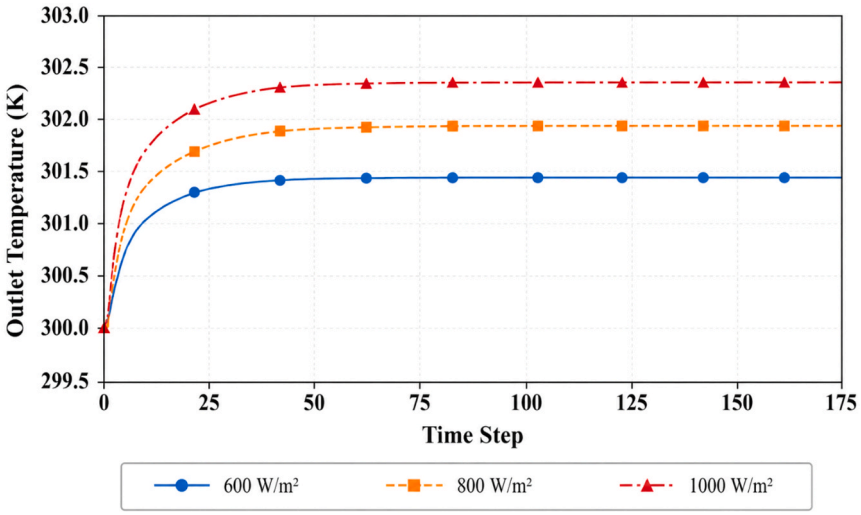
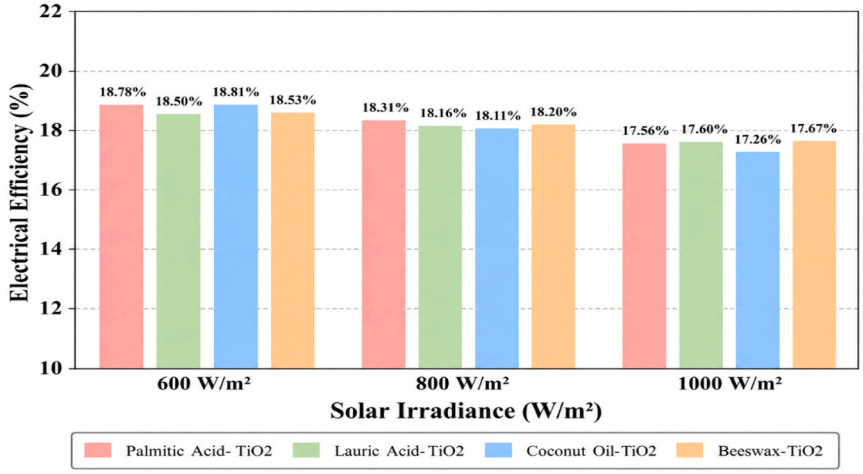
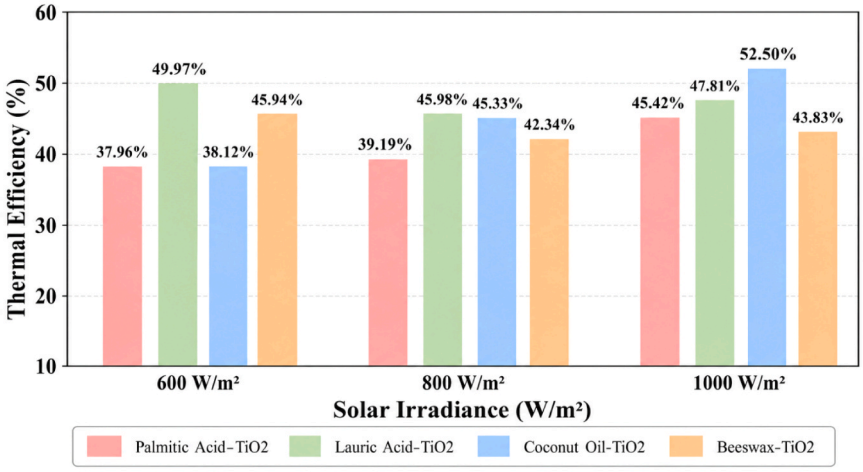


Fig. 10. Outlet temperature distribution of the hybrid PV cooling system under solar irradiation intensities of 600 W/m², 800 W/m², and 1000 W/m².



(a)



(b)

Fig. 11. Electrical efficiency and thermal efficiency performance of the hybrid PV cooling system using nano-enhanced bio-PCM under different solar irradiation intensities: (a) Electrical efficiency and (b) Thermal efficiency.

progression and heat absorption can temporarily moderate the temperature-induced reduction in electrical efficiency. Thus, the electrical response should be interpreted in terms of the transient thermal state of each PCM configuration rather than by irradiation intensity alone, particularly because differences in latent heat utilization and thermal diffusion can yield different efficiency responses at intermediate and high irradiation levels. This behavior indicates that although Coconut Oil-TiO_2 exhibited strong latent heat absorption and rapid melting, excessive thermal accumulation under severe irradiation conditions still led to greater temperature-induced electrical degradation. Furthermore, the relatively stable efficiency reduction observed across all nano-enhanced bio-PCM materials confirms that the combined integration of TiO_2-enhanced PCM and nanofluid cooling successfully minimized excessive PV temperature rise and maintained acceptable electrical conversion performance under varying environmental operating conditions.

The thermal efficiency of the integrated cooling system is illustrated in Fig. 11**(b)**. The results demonstrate that increasing solar irradiance intensity significantly enhances the thermal efficiency of nano-enhanced bio-PCM cooling configurations, as higher solar heat flux intensifies thermal energy absorption and latent heat storage within the PCM layer. Under 600 W/m^2 irradiation, Lauric Acid-TiO_2 achieved the highest thermal efficiency of approximately 49.97%, while Coconut Oil-TiO_2 and Palmitic Acid-TiO_2 exhibited lower efficiencies near 38%. As the irradiation intensity increased to 800 W/m^2, Coconut Oil-TiO_2 exhibited a substantial increase in thermal efficiency, reaching approximately 45.33%, indicating improved thermal absorption and stronger phase-transition activity under elevated thermal loading conditions. At the highest irradiation intensity of 1000 W/m^2, Coconut Oil-TiO_2 achieved a maximum thermal efficiency of approximately 52.50%, which was considerably higher than that of the other investigated nano-enhanced bio-PCM materials. The superior thermal efficiency observed in Coconut Oil-TiO_2 confirms that the material exhibits highly effective latent heat storage and enhanced thermal diffusion characteristics resulting from the incorporation of TiO_2 nanoparticles. In addition, the increasing thermal efficiency trend observed across all cooling configurations demonstrates that the hybrid integration of a nano-enhanced bio-PCM with a TiO_2/water nanofluid successfully improved thermal energy absorption, stabilized transient heat accumulation, and enhanced the overall thermal management capability of the proposed PV cooling system under high solar irradiation conditions.

The statistical significance of the variation in electrical efficiency was further evaluated using ANOVA, as summarized in Table 4. The results indicate that solar irradiation intensity had a statistically significant influence on the electrical efficiency of the hybrid PV cooling system, as evidenced by an F-value of 49.722, substantially higher than the critical F-value of 5.143. In addition, the P-value of 0.000184 remained significantly below the 0.05 threshold, confirming that

Table 4
ANOVA analysis of electrical efficiency under different solar irradiation intensities and nano-enhanced bio-PCM configurations.

Source of Variation	SS	df	MS	F	P-value	F crit
Solar Irradiation Intensities	0.000259522	2	0.000129761	49.72208622	0.000184	5.143253
Nano-enhanced Bio-PCM	4.24917E-06	3	1.41639E-06	0.542735498	0.670652	4.757063
Error	1.56583E-05	6	2.60972E-06	–	–	–
Total	0.000279429	11	–	–	–	–

variations in solar irradiation strongly affected PV electrical conversion performance. This behavior is consistent with previously observed thermal response characteristics: increasing irradiation intensity intensifies thermal accumulation and accelerates PV temperature rise, thereby reducing electrical efficiency. In contrast, the variation in nano-enhanced bio-PCM materials had a comparatively small influence on electrical efficiency, as the F-value of 0.543 remained well below the critical F-value of 4.757. At the same time, the P-value of 0.671 indicated that the variation among the investigated PCM configurations was not statistically significant. These findings demonstrate that although the integration of nano-enhanced bio-PCM successfully stabilized PV thermal behavior and minimized excessive thermal degradation, the dominant factor controlling electrical efficiency remained the magnitude of absorbed solar irradiation. Furthermore, the relatively small error variance obtained in the ANOVA analysis confirms the numerical stability and consistency of the proposed hybrid cooling framework during transient thermal–electrical simulations.

The ANOVA evaluation for thermal efficiency is presented in Table 5. Unlike electrical efficiency, the statistical analysis indicates that neither solar irradiance intensity nor variation in the nano-enhanced bio-PCM had a statistically significant effect on the thermal efficiency response of the integrated cooling system. The F-values obtained for solar irradiation intensity and the nano-enhanced bio-PCM were 1.460 and 1.538, respectively, both of which remained below their respective critical F-values. Moreover, the P-values of 0.304 and 0.299 were substantially higher than the 0.05 significance threshold, indicating that the observed differences in thermal efficiency across operating conditions were not statistically significant. This behavior suggests that the hybrid cooling architecture maintained relatively stable thermal energy absorption performance despite variations in irradiation intensity and PCM composition. The relatively uniform thermal efficiency distribution confirms that the synergistic interaction between latent heat storage and nanofluid-assisted convective cooling successfully stabilized transient heat transfer processes within the PV cooling structure. Therefore, the designation of Coconut Oil-TiO_2 as the superior configuration is based not on thermal efficiency alone, but on its combined transient performance, particularly its consistently lower PV temperature, faster melting progression, higher melting fraction, and stable thermal regulation under the investigated irradiation conditions. These complementary indicators provide a broader basis for evaluating cooling effectiveness and help explain why Coconut Oil-TiO_2 was identified as the most favorable configuration, despite the lack of a statistically significant difference in thermal efficiency among the PCM materials. In addition, the moderate error variance observed in the ANOVA analysis indicates that the proposed thermal management system maintained consistent thermal regulation throughout the investigated operating conditions. Therefore, the statistical findings further strengthen the overall conclusion that the integrated nano-enhanced bio-PCM and TiO_2/water nanofluid cooling framework provided reliable thermal stabilization and maintained balanced thermal management performance under varying environmental thermal loads.

3.5. Hybrid RNN–LSTM prediction performance for PV thermal behavior

Following the stable thermal regulation and efficiency characteristics demonstrated by the hybrid PV cooling configuration, the predictive capability of the implemented hybrid RNN–LSTM architecture was subsequently evaluated to analyze its effectiveness in forecasting transient PV thermal behavior under varying operating conditions. The comparison between actual and predicted PV temperatures is shown in Fig. 12**(a)**. The results indicate that the predicted temperature values closely followed the actual thermal response generated from the CFD simulation dataset throughout the prediction sequence. Most predicted temperatures exhibited only very small deviations from the actual values, confirming that the hybrid RNN–LSTM model successfully captured the nonlinear temporal relationship and transient thermal evolution characteristics of the integrated PV cooling system. The majority of prediction deviations remained below 0.3 K, while several samples exhibited extremely low errors of 0.001 K, indicating excellent prediction precision and a highly stable learning capability during sequential forecasting. In addition, the close agreement between the actual and predicted temperature distributions demonstrates that the implemented temporal feature extraction, using sequential data on PV temperature, outlet temperature, electrical efficiency, and thermal efficiency, effectively improved the model's learning performance. The stable prediction behavior also confirms that the integrated hybrid cooling architecture produced sufficiently consistent thermal patterns, enabling the hybrid RNN–LSTM framework to accurately identify transient thermal trends and dynamic temperature evolution under different irradiation conditions. Therefore, the developed deep learning prediction model demonstrated strong capability for intelligent thermal forecasting and adaptive PV thermal management applications.

The residual and absolute error characteristics generated by the hybrid RNN–LSTM prediction process are illustrated in Fig. 12(b) and Fig. 12**(c)**. The residual distribution exhibited relatively balanced positive and negative fluctuations around zero, indicating that the developed prediction model did not experience significant systematic overestimation or underestimation during thermal forecasting. Most residual values remained within a narrow range below ± 0.3 K, confirming strong numerical consistency and stable prediction reliability across the investigated dataset. Furthermore, the absolute error distribution showed that most errors remained very low, with most samples remaining below 0.2 K despite the transient, nonlinear thermal behavior of the integrated PV cooling system. Although several isolated prediction points produced larger deviations approaching approximately 0.8 K, these values remained relatively small compared with the overall operating temperature range exceeding 300 K. The low prediction error magnitude confirms that the hybrid RNN–LSTM architecture effectively

Table 5
ANOVA analysis of thermal efficiency under different solar irradiation intensities and nano-enhanced bio-PCM configurations.

Source of Variation	SS	df	MS	F	P-value	F crit
Solar Irradiation Intensities	0.004908215	2	0.002454108	1.460415343	0.304255	5.143253
Nano-enhanced Bio-PCM	0.007754303	3	0.002584768	1.53816983	0.298589	4.757063
Error	0.010082505	6	0.001680418	–	–	–
Total	0.022745023	11	–	–	–	–

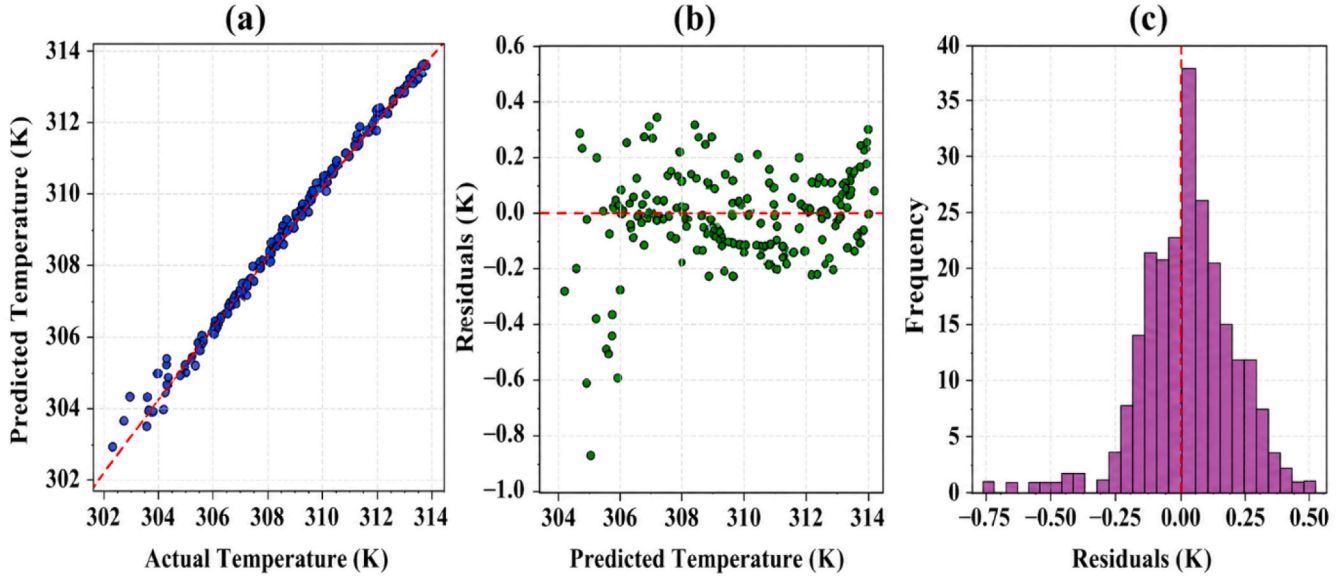


Fig. 12. Hybrid RNN–LSTM prediction performance for transient PV thermal behavior: (a) Comparison between actual and predicted PV temperatures, (b) Residual distribution of temperature prediction, and (c) Absolute error distribution of the hybrid RNN–LSTM model.

minimized temporal learning instability and successfully captured both short-term thermal fluctuations and long-term thermal evolution patterns from the CFD-generated dataset. In addition, the highly stable residual behavior further supports the previous ANOVA and thermal efficiency analyses, which showed that the proposed hybrid cooling configuration demonstrated numerically stable thermal regulation performance under varying irradiation conditions. Consequently, integrating CFD simulation with a hybrid RNN–LSTM model successfully provided an intelligent, adaptive, and highly reliable framework for next-generation PV thermal management and predictive thermal monitoring applications.

The training convergence characteristics of the developed hybrid RNN–LSTM model are shown in Fig. 13**(a)**. The results demonstrate that the implemented deep learning architecture exhibited highly stable convergence throughout training, as indicated by a continuous reduction in loss from approximately 0.000295 in the initial epoch to nearly 0.000201 in the final epoch. This gradual, monotonic decrease in loss confirms that the hybrid RNN–LSTM framework successfully optimized temporal learning parameters and effectively minimized prediction uncertainty during iterative training. Simultaneously, the learning progress curve consistently increased from approximately 0.516–0.982, indicating substantial improvement in the model's learning capability and feature extraction performance as training epochs progressed. The smooth convergence trend observed throughout the learning process further confirms that the integrated CFD-generated thermal dataset provided sufficiently stable temporal patterns for sequential thermal forecasting applications. In addition, the absence of significant oscillatory behavior during training indicates that the proposed hybrid RNN–LSTM architecture maintained strong numerical stability and avoided severe overfitting, despite the PV cooling system's nonlinear transient thermal characteristics. These findings demonstrate that the implemented learning framework possessed highly reliable adaptive learning capability for intelligent PV thermal prediction under dynamically changing thermal operating conditions.

The evolution of prediction accuracy during training is further illustrated in Figs. 13(b) and 13(c), showing continuous learning and a

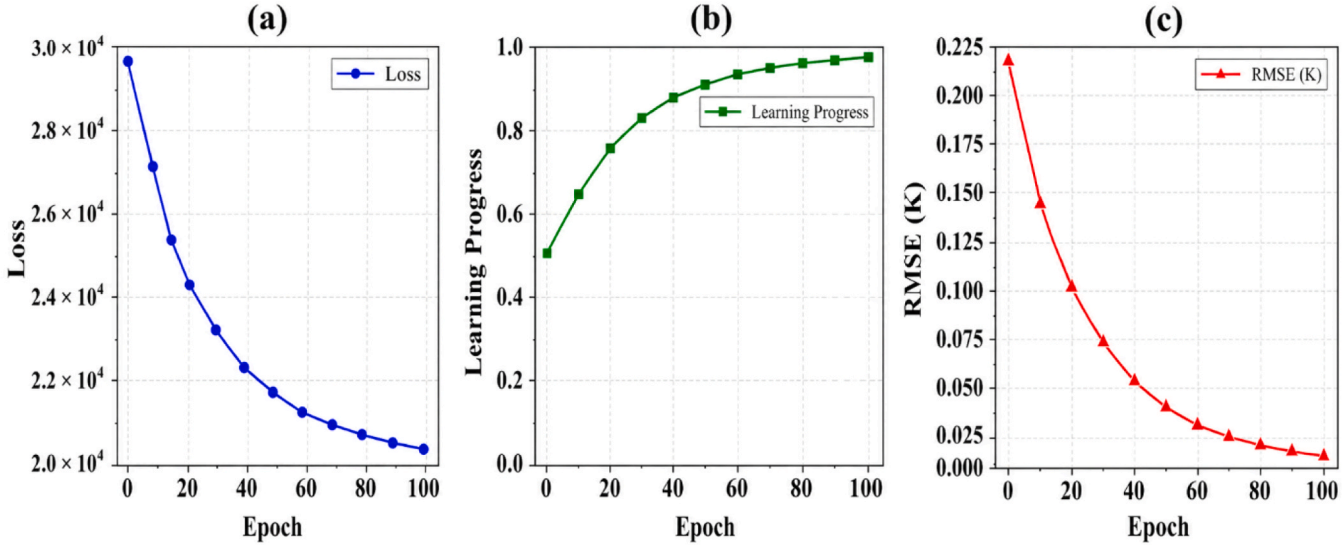


Fig. 13. Training convergence and prediction accuracy performance of the hybrid RNN–LSTM model: (a) Loss convergence during training epochs, (b) Learning progress evolution, and (c) RMSE reduction throughout the training process.

reduction in RMSE. The results indicate that the RMSE decreased gradually from approximately 0.222 K at the initial epoch to nearly 0.021 K at the final epoch, confirming the hybrid RNN–LSTM model's remarkable improvement in prediction precision. The progressive reduction in RMSE demonstrates that the developed learning architecture effectively captured both short-term thermal fluctuations and long-term thermal evolution patterns from the transient CFD simulation dataset. Furthermore, the extremely low final RMSE confirms that the prediction deviation remained very small relative to the overall PV operating temperature range, which exceeds 300 K, thereby indicating highly accurate thermal forecasting capability. The simultaneous increase in learning progress and reduction in RMSE confirm that the model training process converged efficiently, without significant instability or divergence during optimization. This behavior strongly supports the previous residual and absolute error analyses, where the developed prediction framework demonstrated highly consistent and reliable thermal forecasting performance. Therefore, the integration of hybrid RNN–LSTM learning with the nano-enhanced hybrid PV cooling configuration successfully established a robust intelligent prediction framework that supports adaptive thermal regulation, predictive monitoring, and next-generation smart PV thermal management systems.

Building upon the highly stable convergence and prediction accuracy achieved by the hybrid RNN–LSTM framework, the optimization characteristics of the integrated PV thermal management system were further evaluated under varying irradiation and thermal operating conditions. The optimization contour and thermal prediction distribution are shown in Fig. 14**(a)**. The results demonstrate that increasing solar irradiance from 600 W/m^2 to 1000 W/m^2 continuously raised the predicted PV operating temperature from approximately 310 K to 330 K, due to greater thermal energy absorbed within the PV structure. Simultaneously, the optimization analysis revealed that the performance improvement parameter gradually decreased from approximately 0.6 to −0.6 as the thermal operating condition shifted toward higher irradiation intensity and higher temperature. This behavior indicates that the thermal management effectiveness of the cooling configuration progressively worsened under severe thermal loading conditions, despite the integration of a nano-enhanced bio-PCM and a TiO_2/water nanofluid. Nevertheless, the predicted thermal contour remained highly smooth and continuous across the investigated operating range, confirming that the developed hybrid RNN–LSTM architecture successfully captured the nonlinear interactions among irradiation intensity, PV temperature evolution, and transient thermal regulation performance. In addition, the stable contour progression further demonstrates that the integrated CFD–deep learning framework effectively preserved numerical stability and adaptive prediction capability even under dynamically changing thermal environments. These findings confirm that the proposed intelligent thermal prediction framework possesses strong capability for real-time PV thermal optimization and adaptive cooling management applications.

The optimality behavior and system stability characteristics are further illustrated in Figs. 14(b) and 14(c) through the variation in the optimality score across the investigated thermal operating conditions. The obtained optimality score remained consistently high, within a narrow range of approximately 0.921–0.923, across the entire range of irradiation and temperature variations, indicating that the proposed hybrid PV cooling system maintained highly stable optimization performance under increasing thermal stress. The slight decrease in the optimality score observed at higher irradiation intensities suggests that elevated thermal accumulation slightly reduced the overall cooling effectiveness and the system's predictive optimization capability. However, the variation remained extremely small, confirming that the integrated nano-enhanced cooling configuration and hybrid RNN–LSTM learning architecture successfully preserved robust thermal regulation and stable predictive performance under transient operating conditions. Furthermore, the relatively uniform optimality distribution indicates that the proposed intelligent framework effectively balances thermal absorption, latent heat storage, and convective cooling across the investigated thermal range. The stable optimization characteristics observed in the present analysis also strongly support the previous RMSE, residual, and ANOVA evaluations, where the integrated cooling and prediction framework demonstrated highly consistent numerical behavior and reliable thermal management capability. Consequently, the integration between CFD simulation, nano-enhanced thermal regulation, and hybrid RNN–LSTM optimization successfully established an intelligent PV thermal management framework with strong adaptability, high predictive stability, and reliable optimization performance for next-generation smart solar energy systems.

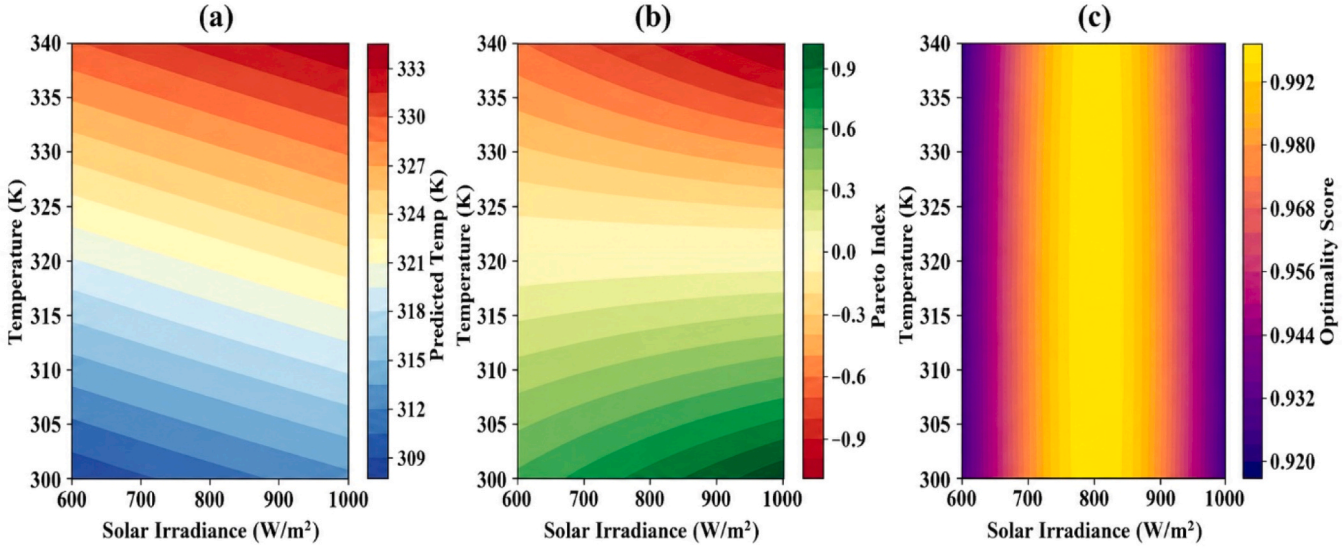


Fig. 14. Optimization contour and intelligent thermal prediction performance of the hybrid PV cooling system: (a) Irradiance–temperature contour prediction, (b) Performance improvement distribution under varying thermal conditions, and (c) Optimality score variation of the hybrid RNN–LSTM optimization framework.

3.6. Sustainability prediction and environmental impact assessment using hybrid RNN–LSTM

Following the successful implementation of intelligent thermal prediction and optimization analysis, the developed hybrid RNN–LSTM framework was further utilized to estimate the long-term sustainability performance and environmental benefits of the proposed hybrid PV cooling system. The predicted environmental impact characteristics are presented in Fig. 15 **(a)**. The results indicate that the nano-enhanced bio-PCM configurations significantly reduced CO_2 emissions by improving PV thermal regulation and enhancing the stability of energy conversion. Among the investigated materials, Palmitic Acid-TiO_2 achieved the highest predicted CO_2 reduction of approximately 126.68 kg/kWp, followed by Lauric Acid-TiO_2 at approximately 119.36 kg/kWp. At the same time, Coconut Oil-TiO_2 and Beeswax-TiO_2 produced lower reductions of approximately 109.83 kg/kWp and 96.57 kg/kWp, respectively. This behavior indicates that the superior thermal stabilization capability of Palmitic Acid-TiO_2 enabled more efficient PV operating conditions, thereby increasing annual electrical energy utilization and reducing carbon-related environmental impacts. Furthermore, the deep learning prediction generated by the hybrid RNN–LSTM architecture confirms that improved transient thermal regulation directly contributed to enhanced sustainability performance under long-term operational conditions. The predicted trends also demonstrate that integrating nano-enhanced bio-PCM and TiO_2/water nanofluid cooling successfully minimizes thermal degradation losses, thereby improving the environmental viability of the proposed PV cooling configuration. In addition, the relatively consistent reduction trends observed across all investigated materials indicate that the developed intelligent prediction framework maintained stable environmental forecasting capability throughout the sustainability evaluation process.

The predicted annual energy- and water-saving performance is further illustrated in Fig. 15(b) and Fig. 15**(c)**. The results demonstrate that Palmitic Acid-TiO_2 achieved the highest predicted annual energy saving of approximately 886.09 kWh/m^2·year and water saving of approximately 2.40 m^3 /year. In comparison, Beeswax-TiO_2 produced the lowest predicted values, approximately 679.58 kWh/m^2·year and 1.84 m^3 /year. The superior sustainability performance observed in Palmitic Acid-TiO_2 confirms that enhanced latent heat storage capability and stable thermal diffusion characteristics significantly improved overall thermal management efficiency and reduced unnecessary energy dissipation during PV operation. Meanwhile, Lauric Acid-TiO_2 and Coconut Oil-TiO_2 also demonstrated substantial predicted sustainability benefits, indicating that all investigated nano-enhanced bio-PCM materials positively contributed to energy conservation and environmental preservation. The hybrid RNN–LSTM model's predictions further verify that intelligent deep learning analysis can effectively estimate long-term environmental performance using transient CFD-based thermal datasets. Moreover, the proportional relationship among CO_2 reduction, annual energy savings, and water conservation confirms that improvements in thermal regulation performance directly enhanced the sustainability characteristics of the integrated PV cooling system. Therefore, the integration between CFD simulation, nano-enhanced thermal regulation, and hybrid RNN–LSTM prediction successfully established a highly adaptive and environmentally sustainable framework for future smart PV thermal management systems and intelligent renewable energy applications.

4. Discussion

4.1. Thermal regulation performance of nano-enhanced bio-PCM and nanofluid integration

The thermal regulation characteristics obtained from the transient CFD simulation demonstrate that integrating a nano-enhanced bio-PCM with a TiO_2/water nanofluid successfully improved the overall thermal stability of the hybrid PV cooling system under varying solar irradiation conditions. Based on the previously presented temperature distribution results, the increase in solar irradiance significantly intensified thermal accumulation within the PV structure, thereby increasing the transient PV operating temperature throughout the simulation duration. Nevertheless, the proposed hybrid cooling configuration was capable of suppressing excessive temperature rise through the synergistic interaction between latent heat absorption and convective heat dissipation. The superior thermal performance observed in Coconut Oil-TiO_2 confirms that the thermophysical properties of the selected bio-PCM strongly influence the effectiveness of transient thermal regulation and heat-diffusion characteristics. In addition, the contour distribution and melting fraction analyses indicate that incorporating TiO_2 nanoparticles accelerated internal thermal diffusion and improved thermal uniformity during the phase transition. Physically, increasing irradiation increases the heat flux imposed on the PV surface, so a larger fraction of the

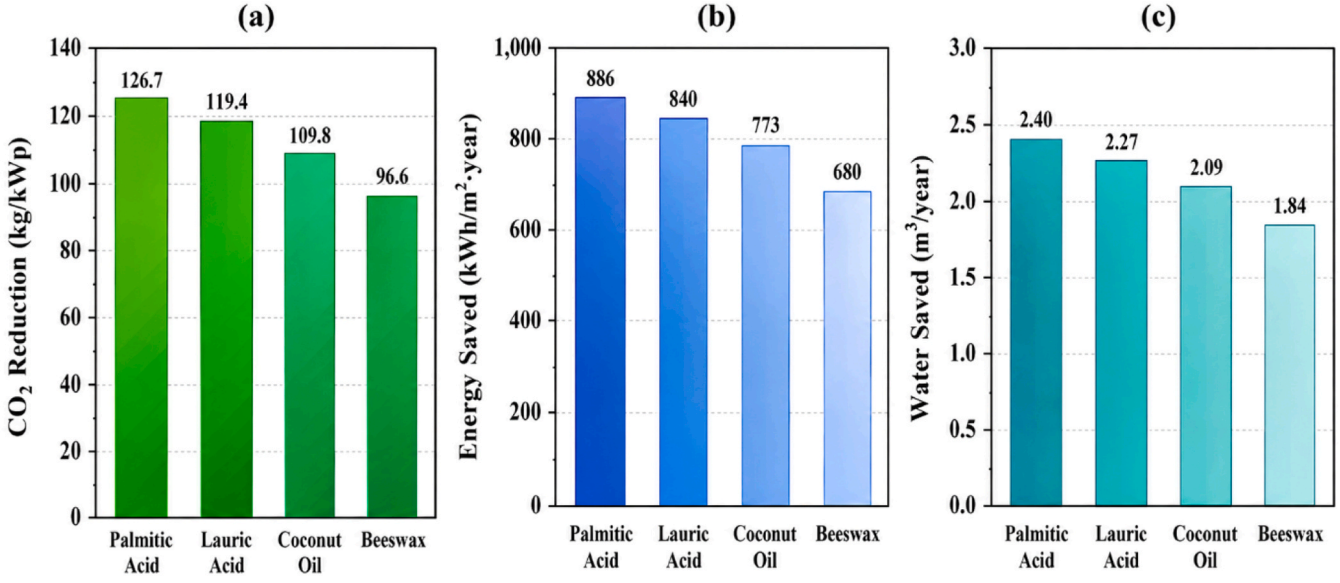


Fig. 15. Hybrid RNN–LSTM sustainability prediction for the proposed hybrid PV cooling system: (a) Predicted CO_2 reduction, (b) Predicted annual energy saving, and (c) Predicted annual water saving for different nano-enhanced bio-PCM configurations.

absorbed energy must be transferred away or stored as latent heat to prevent further temperature escalation. The lower temperature response of Coconut Oil-TiO_2 can therefore be associated with its faster activation of latent heat storage. In contrast, TiO_2-enhanced heat diffusion promotes more effective redistribution of heat from the heated PV region into the cooling medium and PCM domain. These findings demonstrate that the combined integration of nano-enhanced latent heat storage and nanofluid-assisted cooling can effectively enhance thermal management capability and improve the operational stability of next-generation PV systems under dynamically changing environmental conditions.

Furthermore, the thermal efficiency characteristics discussed in the previous section indicate that increasing the irradiation intensity continuously enhanced the integrated cooling system's thermal absorption capability. The stable thermal response observed throughout the transient simulation period confirms that the nano-enhanced bio-PCM's latent heat storage mechanism successfully delayed rapid thermal escalation despite prolonged solar exposure. As summarized in Table 6, Coconut Oil-TiO_2 consistently exhibited the best combined thermal regulation performance, owing to its stable melting behavior, enhanced thermal diffusivity, and higher transient thermal absorption capacity compared with the other investigated materials. Meanwhile, Beeswax-TiO_2 exhibited relatively slower heat diffusion despite its high latent heat storage capacity, indicating that thermal conductivity remained an important factor influencing overall cooling effectiveness. This behavior occurs because latent heat capacity alone does not determine the rate at which a PCM absorbs thermal energy; heat must first be conducted through the PCM matrix to activate the phase transition over a larger volume. Consequently, the slower response of Beeswax-TiO_2 indicates that thermal transport limitations can restrict latent heat utilization even when the material has substantial storage capacity. In contrast, the faster response of Coconut Oil-TiO_2 reflects a more effective balance between heat diffusion and phase-transition absorption. The ANOVA analysis further confirms that although irradiation intensity strongly affected electrical efficiency, the variation in thermal efficiency among the investigated nano-enhanced bio-PCM materials remained statistically stable under the applied operating conditions. Therefore, the proposed hybrid cooling architecture successfully established a balanced interaction between latent heat storage and convective heat transfer, enabling stable PV thermal regulation and improved reliability of long-term thermal management.

4.2. Intelligent prediction capability of the hybrid RNN–LSTM framework

The hybrid RNN–LSTM architecture demonstrated highly stable predictive performance for forecasting the transient thermal behavior of the proposed hybrid PV cooling system. The close agreement between the predicted and actual PV temperature distributions confirms that the implemented deep learning framework successfully captured both short-term thermal fluctuations and long-term thermal evolution patterns generated from the CFD simulation dataset. The extremely low residual and RMSE values obtained during the prediction process indicate that the hybrid learning architecture effectively minimized prediction uncertainty and maintained stable numerical convergence throughout the training and validation stages. In addition, the gradual reduction in training loss and the continuous increase in learning progress demonstrate that the implemented sequential learning mechanism was capable of extracting nonlinear thermal patterns associated with variations in irradiation intensity, PCM melting behavior, and nanofluid cooling interactions. The stable prediction behavior further confirms that the CFD-generated dataset possessed sufficiently consistent temporal characteristics for adaptive thermal forecasting applications. Consequently, integrating a transient CFD simulation with a hybrid RNN–LSTM prediction successfully established an intelligent thermal management framework capable of supporting predictive monitoring and adaptive PV cooling optimization.

The optimization contour analysis additionally demonstrates that the proposed prediction framework maintained highly stable optimality characteristics under varying thermal operating conditions. Although increasing irradiation intensity progressively increased PV operating temperature and reduced performance improvement indicators, the optimality score remained consistently high throughout the investigated thermal range, indicating robust predictive stability and adaptive learning capability. As summarized in Table 7, the hybrid RNN–LSTM architecture achieved excellent thermal forecasting performance, with extremely low prediction deviation and highly stable convergence. The implemented temporal feature extraction mechanism, which uses sequential data on PV temperature, outlet temperature, electrical efficiency, and thermal efficiency, significantly improved learning stability and prediction precision in transient thermal forecasting. Furthermore, the balanced residual distribution around zero confirms that the developed model did not exhibit severe overestimation or underestimation during prediction. However, because the RNN–LSTM model was trained exclusively on CFD-generated data, its predictive reliability has been demonstrated only within the numerical operating domain represented by the simulation dataset. Differences between simulated and experimental PV behavior, including measurement uncertainty, environmental fluctuations, material property variations, and unmodeled heat transfer effects, may affect prediction accuracy when the model is applied to real systems. Therefore, the proposed intelligent framework demonstrates strong potential for real-time thermal prediction, adaptive PV thermal optimization, and intelligent operational control in future smart renewable energy systems.

4.3. Sustainability implications and research limitations

The sustainability prediction results from the hybrid RNN–LSTM framework confirm that integrating a nano-enhanced bio-PCM with TiO_2/water nanofluid cooling provides significant environmental and operational advantages for advanced PV thermal management systems. The predicted CO_2 reduction, annual energy savings, and water-saving performance demonstrate that improved thermal regulation directly contributes to enhanced sustainable energy use and reduced

Table 6
Comparative thermal regulation characteristics of nano-enhanced bio-PCM materials under varying irradiation conditions.

Nano-enhanced bio-PCM	Dominant Thermal Characteristic	Thermal Regulation Performance	Melting Stability	Overall Cooling Capability
Palmitic Acid-TiO_2	Stable thermal diffusion	Moderate	Stable	Good
Lauric Acid-TiO_2	Slower thermal response	Moderate–Low	Moderate	Fair
Coconut Oil-TiO_2	Rapid latent heat absorption	Very High	Excellent	Excellent
Beeswax-TiO_2	High latent heat storage	Moderate	Stable but slower	Good

Table 7
Prediction performance characteristics of the hybrid RNN–LSTM framework.

Prediction Parameter	Observed Performance	Prediction Stability
Temperature prediction accuracy	Very high	Stable
RMSE behavior	Continuously decreasing	Excellent
Residual distribution	Balanced near zero	Highly stable
Loss convergence	Smooth and monotonic	Strong convergence
Adaptive learning capability	Effective under varying irradiation	Robust
Thermal forecasting reliability	Highly consistent	Excellent

environmental impact. Among the investigated materials, Palmitic Acid-TiO_2 achieved the highest predicted sustainability performance due to its stable thermal diffusion behavior and improved thermal management during long-term operation. Meanwhile, Coconut Oil-TiO_2 exhibited excellent transient thermal absorption, thereby improving thermal efficiency under high-irradiation conditions. The proportional relationship between energy-saving performance and CO_2 reduction further confirms that enhanced thermal stabilization can improve PV operational efficiency and reduce energy losses from excessive thermal accumulation. As summarized in Table 8, the proposed hybrid PV cooling framework demonstrates strong potential to support future environmentally sustainable and intelligent renewable energy systems through adaptive thermal management and predictive optimization.

Despite the promising results obtained in this study, several research limitations should be acknowledged to inform future improvements and experimental validation. First, the present investigation was conducted entirely using transient CFD simulations and deep learning predictions, without direct experimental testing under real environmental operating conditions. Consequently, several practical factors, such as outdoor weather fluctuations, material degradation, flow instability, and long-term nanoparticle sedimentation, were not fully accounted for in the numerical framework. Second, the thermophysical properties of the nano-enhanced bio-PCM and the nanofluid were assumed to remain constant throughout the simulation. In contrast, actual operating conditions may produce temperature-dependent variations in properties that influence transient heat transfer behavior. This assumption may introduce prediction errors during the phase-transition region because thermal conductivity, specific heat capacity, density, and latent-heat-related behavior can vary with temperature and phase fraction, thereby altering the rate and spatial distribution of heat absorption. Incorporating temperature- and phase-dependent properties obtained from material characterization would therefore improve the representation of transient heat transfer and increase the reliability of the predicted PV temperature and melting behavior. Third, the implemented hybrid RNN–LSTM framework used CFD-generated datasets with limited irradiation variation, which may constrain the model's generalization under highly dynamic climatic conditions. Furthermore, economic feasibility analysis, lifecycle assessment, and long-term maintenance evaluation were not comprehensively investigated in the present work. To experimentally validate the numerical predictions, a controlled PV cooling test should be conducted using the same PV geometry, bio-PCM configurations, TiO_2/water nanofluid concentration, irradiation levels, and operating duration as the CFD model, with repeated measurements performed under equivalent boundary conditions. The most critical parameters to measure are PV surface temperature, PCM temperature and melting fraction, nanofluid inlet and outlet temperature, flow rate, electrical power or electrical efficiency, and thermal efficiency, because these measurements directly correspond to the principal numerical outputs and allow quantitative validation of both thermal and electrical responses. Therefore, future studies should incorporate experimental validation, a broader range of environmental operating conditions, advanced optimization algorithms, and comprehensive techno-economic analyses to improve the practical applicability and industrial feasibility of the proposed intelligent PV thermal management framework.

Table 8
Sustainability implications and research limitations of the proposed hybrid PV cooling framework.

Aspect	Main Findings	Implication
CO_2 reduction	Improved through stable thermal regulation	Better environmental sustainability
Annual energy saving	Increased due to improved PV thermal stability	Enhanced operational efficiency
Water saving	Reduced cooling-related resource demand	Improved resource sustainability
Numerical simulation limitation	No direct experimental validation	Requires controlled experimental validation using equivalent operating conditions
Material assumption limitation	Constant thermophysical properties assumed	Requires temperature- and phase-dependent property characterization and implementation
Prediction dataset limitation	Limited operating condition variation	Requires broader dataset integration
Experimental validation protocol	PV temperature, PCM temperature, and melting fraction, nanofluid inlet/outlet temperature, flow rate, electrical efficiency, and thermal efficiency should be measured under CFD-equivalent conditions	Enables quantitative validation of thermal–electrical numerical predictions

5. Conclusion

This study established an integrated framework combining nano-enhanced bio-PCM, TiO_2/water nanofluid cooling, CFD simulation, and hybrid RNN–LSTM prediction for intelligent PV thermal management under varying solar irradiation conditions. The key contributions and principal findings can be summarized as follows: (1) Coconut Oil-TiO_2 provided the strongest transient thermal regulation, characterized by lower PV operating temperature, faster and more stable melting progression, and effective thermal absorption under elevated irradiation; (2) TiO_2 incorporation enhanced thermal diffusion and promoted more uniform heat distribution within the PCM layer; (3) increasing solar irradiation accelerated PCM melting and increased thermal energy absorption, while simultaneously increasing thermal stress on the PV system; and (4) the electrical and thermal responses exhibited different trends, with electrical efficiency being more sensitive to irradiation-induced temperature changes, whereas thermal efficiency remained comparatively stable across the investigated configurations. The ANOVA results further demonstrated that solar irradiation significantly affected electrical efficiency. In contrast, variations in nano-enhanced bio-PCM did not produce statistically significant differences in thermal efficiency, indicating that PCM superiority should be evaluated using multiple thermal indicators rather than thermal efficiency alone. These findings confirm that the combined latent heat storage and TiO_2/water nanofluid cooling mechanisms provide an effective approach for controlling transient thermal accumulation and improving the thermal reliability of PV systems.

The second major contribution is the development of an intelligent prediction framework based on a hybrid RNN–LSTM model to capture the nonlinear, transient thermal behavior of the hybrid PV cooling system. The prediction results demonstrated strong agreement between actual and predicted PV temperatures, as evidenced by stable convergence, low residual distributions, and a continuously decreasing RMSE during the prediction process. The integration of CFD-generated thermal data with an RNN–LSTM therefore provides a predictive layer that captures the dynamic relationships among solar irradiation, thermal accumulation, PCM melting, and PV temperature evolution, thereby extending the proposed system beyond conventional simulation-based thermal assessment. The sustainability analysis further showed that the proposed framework can contribute to reduced CO_2 emissions, annual energy savings, and water resource demand. However, Palmitic Acid-TiO_2 produced the highest predicted sustainability performance, while Coconut Oil-TiO_2 provided the strongest transient thermal regulation. Collectively, these findings demonstrate that the proposed framework contributes not only to PV temperature reduction but also to intelligent prediction and sustainability-oriented thermal management, providing a foundation for future adaptive PV cooling systems. Nevertheless, the numerical nature of this study means that the reported performance should not be interpreted as evidence of long-term

durability, operational reliability, or real-time field adaptability. In particular, the use of constant thermophysical properties, CFD-generated datasets, and controlled irradiation conditions limits the present framework's ability to represent material degradation, nanoparticle sedimentation, environmental fluctuations, flow instability, and other time-dependent field effects.

Several limitations should nevertheless be addressed before practical deployment. The present study relies on transient CFD simulations and CFD-generated datasets, with no direct experimental validation, and assumes constant thermophysical properties and a limited range of irradiation conditions. Future work should therefore prioritize controlled experimental validation using the same PV geometry, PCM configurations, TiO_2/water nanofluid conditions, irradiation levels, and operating durations used in the numerical model, with simultaneous measurement of PV temperature, PCM temperature, and melting fraction, nanofluid inlet and outlet temperature, flow rate, electrical efficiency, and thermal efficiency. Future investigations should also incorporate temperature- and phase-dependent thermophysical properties, a broader range of climatic conditions, larger experimental datasets, and advanced optimization algorithms to improve model generalization and predictive robustness. Comprehensive techno-economic analysis, lifecycle assessment, long-term material stability, nanoparticle sedimentation, and system maintenance should additionally be evaluated to determine the practical and industrial feasibility of the proposed intelligent hybrid PV cooling architecture. These developments would enable progression from numerical and predictive validation toward experimentally verified, optimized, and scalable intelligent PV thermal management systems.

CRediT authorship contribution statement

Singgih Dwi Prasetyo: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Zainal Arifin:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yuki Trisnoaji:** Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Mohd Afzanizam Mohd Rosli:** Writing – review & editing, Writing – original draft, Supervision, Software, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ahmad Fudholi:** Writing – review & editing, Writing – original draft, Supervision, Software, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethical approval

Not applicable.

Consent to participate

Not applicable.

Consent to publish

Not applicable.

Human and animal rights

The authors declare that the work described has not involved experimentation on humans or animals.

Informed consent and patient details

The authors declare that the work described does not involve patients or volunteers.

Clinical trial number

Not applicable.

Declaration of Generative AI and AI-assisted technologies in the writing process

Artificial intelligence tools were not used to generate, analyze, interpret, or modify the scientific results, numerical simulations, machine learning models, design optimization procedures, or conclusions presented in this study. **Grammarly** was used solely for English grammar checking, language proofreading, and improvement of linguistic clarity. The authors remain fully responsible for the scientific content, interpretation, accuracy, and final wording of the manuscript.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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